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Energy, economic and environmental impacts of cryptocurrency mining: a review of sustainable integration pathways in power systems

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A R T I C L E I N F O

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A B S T R A C T

The integration of cryptocurrency mining into modern power systems is increasingly viewed as both a challenge and an opportunity. Proof of Work (PoW) mining consumes 100–130 TWh annually, generates 48.6–64.4 Mt of CO_2, and creates significant electronic waste. Unlike conventional loads, mining facilities are highly flexible; they can curtail demand within minutes and join demand response programs, making them valuable for grid stability and renewable energy integration. This review examines cryptocurrency mining within electric grids across technical, economic, environmental, and policy dimensions. It explores facility operations, their role in renewable adoption and storage management, and their capacity to provide ancillary services that enhance grid flexibility, while also addressing environmental and regulatory implications. A systematic review methodology was applied, using keyword based searches in leading engineering databases and comparative analysis of quantitative results with real-world projects. Evidence indicates that coordinated mining can reduce microgrid operational costs by up to 46 % and improve distributed solar economics by more than 60 %, whereas unmanaged growth risks higher emissions and infrastructure stress. Key gaps persist, including a lack of standardized integration protocols, limited stability assessments, and weak policy frameworks. A multidisciplinary framework is proposed to guide technical, economic, environmental, and regulatory integration.

Introduction

The global electric power system is undergoing a profound transformation, shifting from conventional centralized power systems to decentralized networks under digital control. Moreover, distributed renewable energy sources, bidirectional power flows, and increasing demands for operational flexibility are being incorporated into modern grids. Consequently, novel approaches are required to ensure system reliability and stability in an increasingly complex energy landscape.

Simultaneously, blockchain-based cryptocurrencies have emerged as both technological innovations and economic phenomena. These digital assets are secured through consensus mechanisms, primarily PoW and Proof of Stake (PoS), which are used to validate transactions across the network. For instance, PoW based cryptocurrencies, such as Bitcoin, require extensive computational effort through mining, in which specialized hardware is employed to solve cryptographic puzzles, confirm transactions, and maintain the blockchain ledger. As a result, global PoW mining operations consume approximately 100 to 130 TWh annually, surpassing the total electricity consumption of countries such as the Netherlands or Argentina [1]. This immense energy demand is associated with substantial environmental impacts, with carbon emissions estimated between 48.6 and 64.4 Mt of CO_2, annually, thus presenting significant challenges to global decarbonization objectives [2].

However, unlike conventional industrial loads, cryptocurrency mining exhibits unique operational characteristics. For example, these operations demonstrate exceptional flexibility through instantaneous load disconnection, geographical independence, and rapid response to grid signals [3]. Moreover, rather than acting solely as a burden, mining can function as a valuable grid asset when strategically integrated with renewable energy infrastructure [4]. In addition, the inherent flexibility of these digital loads allows them to serve as demand response resources, providing ancillary services and absorbing excess renewable generation during periods of oversupply. Fig. 1 illustrates a comprehensive framework for integrating cryptocurrency mining within modern power systems, encompassing transmission, distribution, and generation levels alongside renewable resources, storage systems, and ancillary service provision.

Renewable energy deployment is projected to exceed 50 percent of

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global electricity generation by 2030 [5]. While offering substantial environmental and economic benefits, the intermittent and stochastic nature of these resources introduces operational challenges for power system stability. These challenges include reduced system inertia, increased frequency and voltage deviations, and curtailment of renewable generation during peak periods [6]. Consequently, by coordinating cryptocurrency mining with renewable energy resources, it is possible to convert excess generation into economic value, effectively creating an "economic battery" a mechanism that requires no physical infrastructure, experiences no degradation, and offers virtually unlimited cycling capability [7].

As the power system evolves, ancillary services are becoming increasingly critical for maintaining stability and reliability, particularly as conventional synchronous generation resources are phased out. For instance, Saleem and Saha [8] show that declining system inertia, coupled with higher penetration of variable renewable energy sources, leads to more frequent and severe grid disturbances, thereby emphasizing the need for fast-responding and flexible support mechanisms. Moreover, cryptocurrency mining facilities, owing to their ability to rapidly adjust power consumption, can operate as controllable loads that provide essential grid services, including frequency regulation, voltage support, and operational reserves [9].

However, current research on cryptocurrency mining within power systems remains fragmented, which limits understanding to isolated technical or economic aspects rather than comprehensive interdisciplinary approaches. In particular, existing literature can be broadly categorized into four primary domains, each characterized by distinct strengths and limitations.

Economic and optimization studies

Research in this domain has primarily concentrated on economic optimization frameworks at the microgrid and distribution levels. Hajipour et al. [10] proposed a stochastic optimization framework illustrating the significant economic potential of cryptocurrency mining as an investment avenue for system operators, while this study demonstrates how strategic load management can enhance system profitability, including the integration of mining operations with local microgrids to optimize electricity costs and improve operational efficiency, and further exploring electricity market participation, whereby subsequent work [11] examined how mining operations can engage with various market mechanisms. These economic analyses primarily focus on local-scale impacts and short-term profitability metrics. Yet, they often overlook broader implications for transmission level system stability and long term market dynamics. In addition, comprehensive risk assessment, including cryptocurrency price volatility, is generally not considered.

Technical flexibility and grid integration studies

Research has also examined the technical flexibility of mining operations and their ability to provide ancillary services within power systems. Abid et al. [12] notably reframed cryptocurrency mining from a passive consumer into an active grid resource, capable of supporting frequency regulation, voltage control, and operating reserves. Further studies [13] confirmed the rapid response and controllability of mining loads. Despite these insights, most research focuses on technical potential and theoretical capabilities, while practical implementation challenges, economic risks due to cryptocurrency market volatility, and comprehensive environmental lifecycle assessments remain insufficiently addressed.

Environmental impact assessments

Environmental research has systematically classified and quantified the ecological concerns associated with cryptocurrency mining. Auer et al. [14] and related studies [15] provided a holistic assessment by evaluating carbon footprints, lifecycle effects, and electronic waste, while examining energy consumption patterns and hardware disposal practices. Additionally, the assessment of renewable energy use and broader sustainability metrics in studies [16,17] has highlighted key environmental challenges and established critical baseline metrics for impact evaluation. Namely, carbon emissions and resource utilization emerge as major concerns, and some analyses specifically quantified emissions under different operational strategies. Modelling scenarios further illustrated potential reductions in emissions when mining loads are flexibly coordinated with renewable generation. These studies particularly underscore the importance of systematic environmental

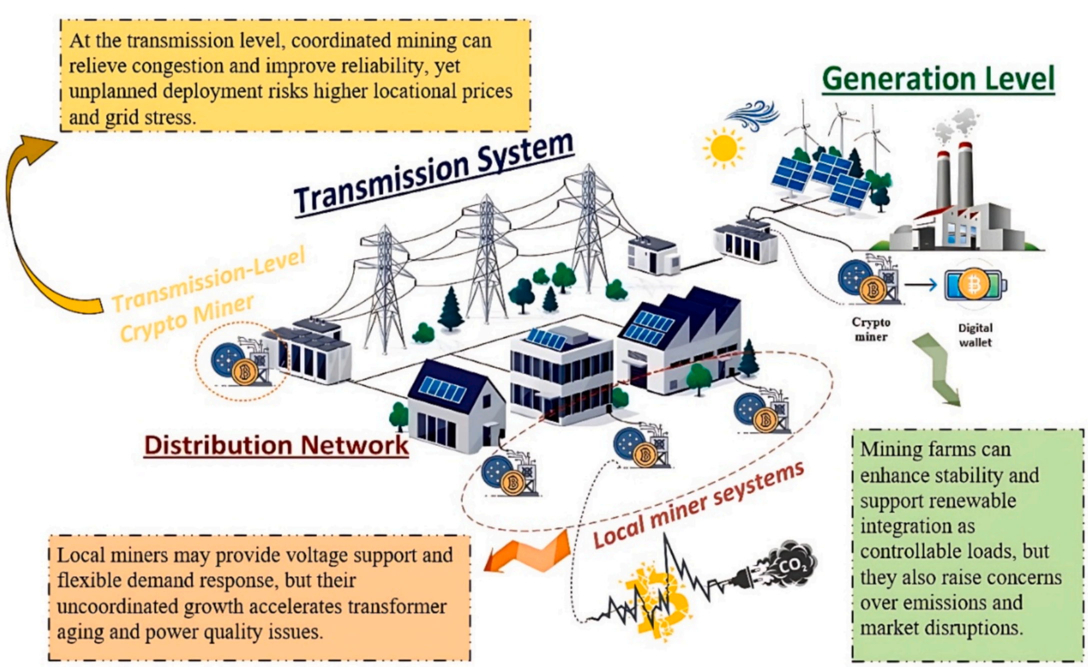

Fig. 1. Multi-level challenges of cryptocurrency mining integration in power systems.

monitoring. Nevertheless, many analyses adopt a predominantly negative perspective, often overlooking the potential for strategically deployed, high flexibility mining loads to support renewable integration, reduce curtailment, and contribute to grid decarbonization when properly managed.

Policy and regulatory frameworks

A limited but growing body of research has begun examining policy implications and regulatory frameworks for integrating cryptocurrency mining with power systems. Recent studies [18] have addressed regulatory compliance, policy design, and governance structures for mining operations. Nevertheless, this area remains underdeveloped, particularly in terms of comprehensive policy frameworks that balance economic opportunities with environmental and grid stability considerations.

Research gap analysis

The fragmented nature of existing research results in several critical knowledge gaps. First, economic optimization models are rarely integrated with environmental impact assessments, producing solutions that may be economically attractive yet environmentally problematic, or vice versa. Second, technical studies of grid integration often overlook realistic economic constraints and market dynamics, limiting practical applicability. Third, policy analyses frequently lack a solid technical foundation to guide effective regulatory design. Consequently, most studies do not offer integrated assessment frameworks that simultaneously account for technical feasibility, economic viability, environmental sustainability, and policy requirements.

To systematically illustrate these research gaps and highlight the comprehensive scope of this review, Table 1 presents a detailed comparison of coverage areas across previous literature, positioning the current study within this research landscape. Building on this, the paper provides a comprehensive, multidisciplinary analysis of PoW based cryptocurrency mining integration with electrical power infrastructure. The study examines technical and economic aspects in detail, focusing on four primary domains: (1) the use of cryptocurrency mining as flexible energy storage; (2) the provision of ancillary services across multiple power system hierarchies; (3) environmental impact assessment and mitigation strategies; and (4) policy frameworks that shape mining integration patterns. Furthermore, insights from major global implementation projects are incorporated to provide practical perspectives on real world deployment scenarios.

This methodological framework is designed to enable a thorough evaluation of cryptocurrency mining's role in modern power systems while highlighting potential pathways for future development. The review follows a systematic approach to identify, analyze, and synthesize scientific literature, with a particular focus on peer reviewed publications from the past five years. Quantitative analyses and real-world case studies are integrated to ensure both contemporary relevance and practical applicability. By systematically categorizing challenges, opportunities, and research directions, this work provides a coherent framework that is expected to be utilized to inform researchers, policymakers, and industry stakeholders.

Specifically, the paper is structured as follows: Section II introduces the technical fundamentals of cryptocurrencies, consensus algorithms, and mining operations. Section III discusses key aspects of modern power systems, including renewable energy integration, energy storage technologies, and ancillary service provision. Section IV forms the core analysis, offering a comprehensive examination of interactions between cryptocurrency mining and the power grid across multiple dimensions. Section V highlights existing challenges and outlines future research priorities, while Section VI presents concluding remarks and strategic recommendations. This section is intended to introduce the fundamental concepts of digital currencies and the technologies underpinning them. It begins with a concise overview of cryptocurrencies and compares the main types, highlighting why Bitcoin has emerged as the most prominent and influential digital currency. The discussion then examines Bitcoin's mining process, including the required hardware and the associated energy consumption challenges. Specifically, the energy-intensive nature of Proof of Work mining is considered a critical factor in both economic and environmental evaluations.

Overview of cryptocurrencies and blockchain technology

This section introduces the fundamental concepts of digital currencies and their underlying technology. It starts with a brief overview of cryptocurrencies and compares their main types, highlighting the prominence and influence of Bitcoin. Building on this, the discussion examines Bitcoin's mining process, detailing the necessary hardware and emphasizing the critical issue of energy consumption.

Fundamentals of cryptocurrencies and blockchain

Cryptocurrencies represent a form of digital money that relies on cryptography to secure transactions and to regulate the creation of new units. As described by Narayanan et al. [19], the concept was developed from decades of research aimed at creating a decentralized financial system independent of central institutions, such as banks. The historical evolution of these currencies is summarized in Fig. 2, illustrating key milestones in their development.

Blockchain technology underpins all cryptocurrencies. Casino et al. [21] define it as a distributed, immutable ledger that records transactions in a chain of blocks. Each block contains a set of transactions, a timestamp, and a cryptographic hash of the preceding block. This structure renders the alteration of past data nearly impossible without controlling the majority of the network's computing power, a task that is considered both economically and technically infeasible [21]. Unlike traditional financial systems, which rely on intermediaries such as banks, blockchain operates on a peer to peer (P2P) basis without a central authority. Song et al. [22] emphasize that all network

Table 1
Comparative analysis of research coverage in previous literature.

Article	Technical infrastructure assessment	Economic & market analysis	Environmental & sustainability metrics	Policy & regulatory analysis	Integrated multidisciplinary approach	Strategic future recommendations
[10]	✓	✓	×	×	×	×
[11]	×	✓	×	×	×	×
[12]	✓	×	×	×	×	×
[13]	✓	×	×	×	×	×
[14]	×	×	✓	×	×	×
[15]	×	×	✓	×	×	×
[16]	×	×	✓	×	×	×
[17]	×	×	✓	×	×	×
[18]	✓	×	×	✓	×	×
This Review	✓	✓	✓	✓	✓	✓

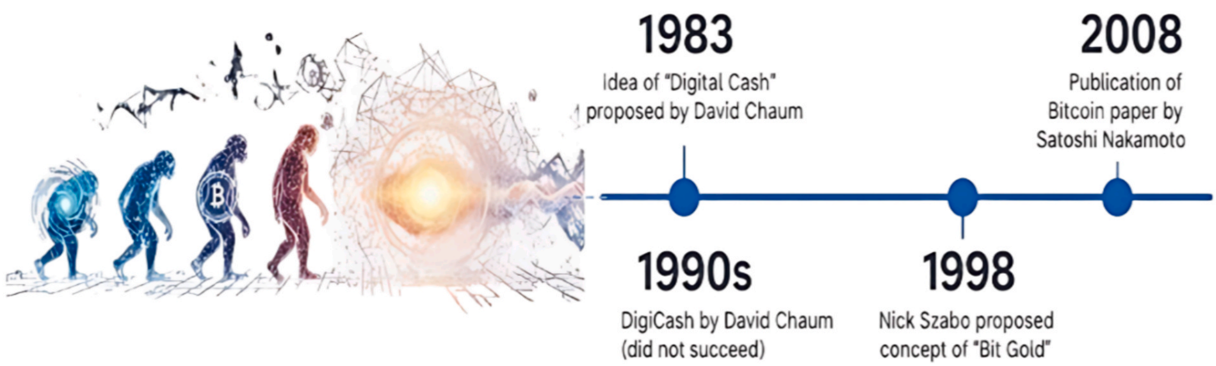

Fig. 2. A timeline of the evolution of decentralized digital currency [19,20].

participants, known as nodes, maintain a copy of the ledger, and any updates are required to achieve consensus across the entire network.

Blockchains can be classified into three primary types based on access permissions and governance structure, as outlined by Zheng et al. [23] Public blockchains, such as Bitcoin and Ethereum, are open to participation and validation by anyone. Private blockchains restrict access to authorized members, typically for enterprise applications. Consortium blockchains are jointly managed by a selected group of organizations, making them suitable for inter-organizational use cases, including supply chain management.

The architecture of a blockchain dictates how participants are able to reach agreement on transaction validity. This process, widely referred to as the consensus mechanism, is critical for overall network security, as highlighted in surveys like Xiao et al. [24] Nakamoto [25] introduced the PoW model, in which miners are required to compete through intensive computation. In contrast, PoS selects validators based on the amount of cryptocurrency they hold and stake, providing a more energy-efficient alternative [26]. Beyond consensus mechanisms, cryptocurrencies can also be categorized by their functional and technical attributes, as outlined in surveys such as Li et al. [27]. Specifically, Coins like Bitcoin (BTC) and Ethereum (ETH) operate on their own independent blockchains. In contrast, Tokens are developed on existing blockchains, for instance, ERC-20 tokens on the Ethereum network. To mitigate price volatility, Stablecoins such as USDT and USDC are pegged to stable assets. Moreover, privacy focused cryptocurrencies, including Monero (XMR) and Zcash, have emerged to prioritize transaction confidentiality [28]. Subsequently, a comparative overview of these categories is presented in Table 2.

Bitcoin: dominance and market position

Bitcoin, introduced by Nakamoto [25], is widely recognized as the first successful public blockchain, earning global trust and market recognition [23]. Its security is grounded in the PoW consensus mechanism, which operates over a highly distributed network. As Stoll et al. [33]observe, this architecture renders coordinated attacks economically and technically unfeasible. Studies such as [34] highlight that Bitcoin's decentralized design, rapid settlement capabilities, and low cross-border transaction costs are considered key factors contributing to its reputation as "digital gold." Specifically, from a market perspective, cryptocurrencies are primarily categorized by their consensus mechanisms into PoW based and PoS based systems. Fig. 3 illustrates that Bitcoin is commanding a 51.2 % market share among PoW cryptocurrencies, with Ethereum, after transitioning to PoS, being held at 8.7 % and ranking second. This dominance underscores Bitcoin's position relative to other altcoins, thereby establishing it as the primary focus of subsequent sections that examine its technological and economic significance.

Bitcoin mining and energy consumption

Building on Bitcoin's market prominence as described in Section 2.2, the following subsection is designed to examine its mining process and the substantial energy requirements associated with cryptocurrency extraction, highlighting the implications for both operational efficiency and environmental impact.

Mining mechanism and consensus protocol

In the PoW system, miners compete to solve a challenging mathematical puzzle by collecting unconfirmed transactions into a block and repeatedly hashing the block's header while modifying a random number called a "nonce." The goal is to find a hash lower than the target set by the network [33]. Once a miner finds a valid hash, it is immediately shared with the network, and other nodes quickly verify its correctness. If confirmed, the new block is added to the blockchain, and the winning miner receives a reward consisting of newly minted Bitcoins and transaction fees, known as the "Coinbase transaction." Fig. 4

Table 2
A comparative analysis of the features, advantages, and disadvantages of major cryptocurrencies.

Cryptocurrency	Features	Advantages	Disadvantages	Ref.
Bitcoin (BTC)		- High security and decentralization - Widespread global adoption - Limited supply (**21** million units)	- High energy consumption - High transaction fees- Low processing speed	[25,29]
Ethereum (ETH)	A platform for smart contracts, transitioning from PoW to PoS	- Supports smart contracts - Extensive development of DApps- Faster transaction speeds compared to Bitcoin	- High gas fees- Limited scalability	[30]
Ripple (XRP)	A global payment system for banks, utilizes a unique consensus algorithm	- High speed (transactions within seconds) - Low transaction fees- Collaboration with banks	- Controlled by Ripple Labs - Not fully decentralized- Legal challenges	[31,32]
Litecoin (LTC)	A Bitcoin fork using the Scrypt algorithm, higher transaction speed	- Lower transaction fees compared to Bitcoin - Faster mining- Better scalability	- Lower adoption than Bitcoin- Competition with other cryptocurrencies	[26]
Stablecoins (USDT, USDC, DAI)	Cryptocurrencies pegged to physical assets such as the US Dollar	- Reduced price volatility - Suitable for daily transactions- Used in decentralized finance (DeFi)	- Trust required in the issuer - Possibility of central control- Legal challenges	[29]

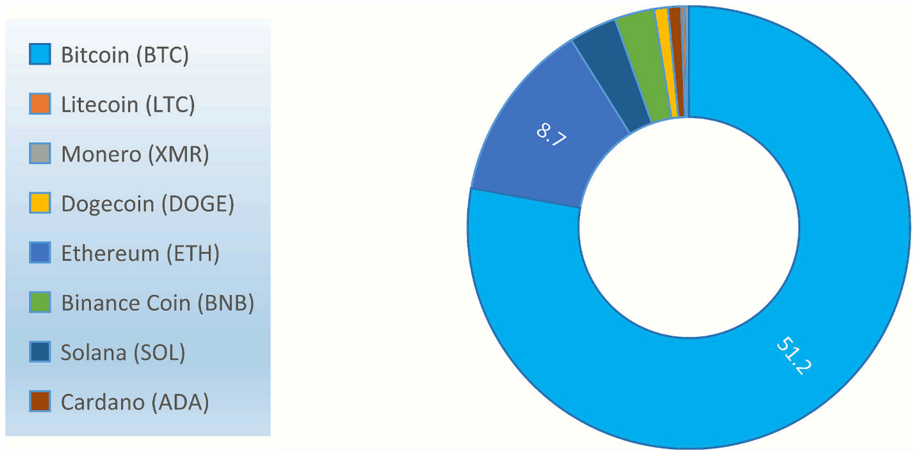

Fig. 3. Market share (%) of major POW & POS cryptocurrencies as of May 26, 2025, highlighting the dominance of Bitcoin and Ethereum in their respective consensus mechanisms. Data sources: [35,36].

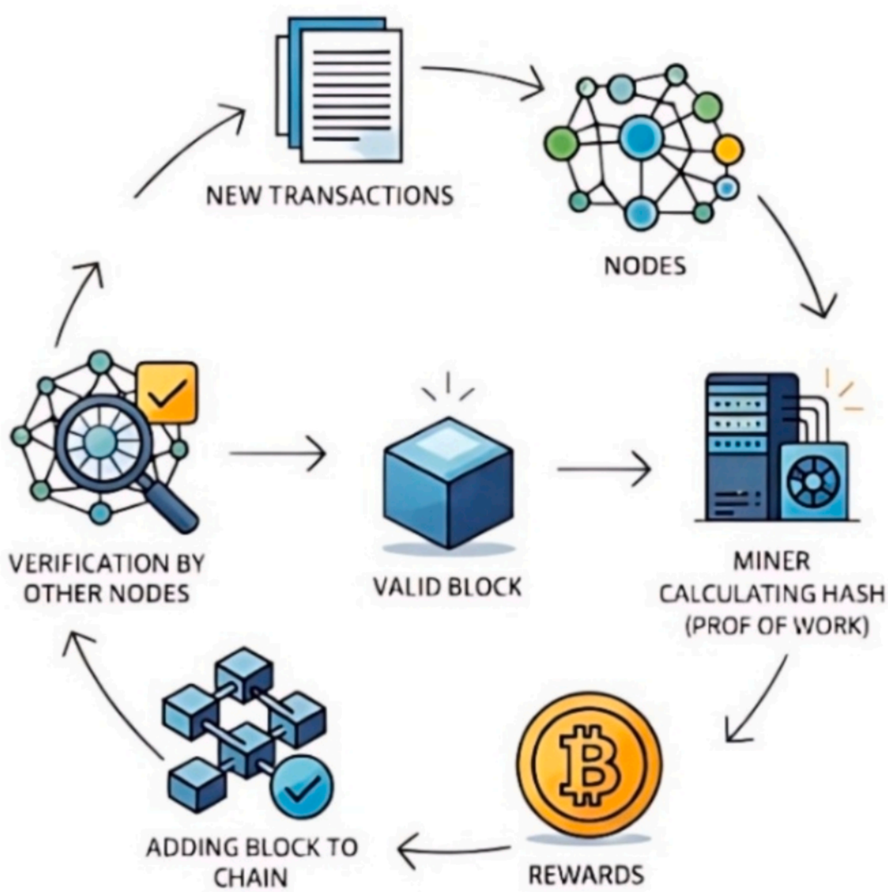

Fig. 4. Flowchart of the Bitcoin (PoW) mining process.

illustrates this mining process. Importantly, as Stoll et al. [33] highlight, altering the blockchain ledger would require control of more than 50 % of the network's total hash power, rendering a "51 % attack" economically impractical. The computational intensity of the PoW process has driven significant advancements in mining hardware, as discussed in the following section.

Mining hardware evolution and operational scale

The high computational demands of Proof of Work have led to specialized hardware development. Mining has progressed from general-purpose CPUs to GPUs, then to FPGAs, and ultimately to ASICs optimized for SHA-256 [37,38]. This evolution increased both processing speed and energy efficiency, but it also reduced the flexibility to mine alternative algorithms. The trade-off between efficiency and adaptability is summarized in Table 3.

Bitcoin miners can be informally categorized based on operational scale: small-scale (S) operations are defined as consuming less than 0.1 MW, medium-scale (M) being classified as consuming up to 1 MW, and

Table 3
The performance vs. flexibility trade-off in cryptocurrency mining hardware.

Hardware type	Example & algorithm	Hash rate	Power consumption	Ref.
CPU	AMD Ryzen 9 5950X (RandomX)	**∼15 KH/s**	**∼140 W**	[37]
GPU	Nvidia RTX 3080 (Ethash)	**∼98 MH/s**	**∼240 W**	[39]
FPGA	Xilinx Virtex-6 (SHA-256)	**∼800 MH/s**	**∼50 W**	[40]
ASIC	Bitmain Antminer S21 Hydro (SHA-256)	**335 TH/s**	**5360 W**	[38]

large-scale (L) exceeding 1 MW [33]. Additionally, to mitigate fluctuations in mining rewards, many miners participate in "mining pools," where computational resources are combined and rewards are shared proportionally.

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Energy consumption models of bitcoin mining

Estimating the total energy consumption of the Bitcoin network without relying on potentially unreliable self-reported data requires the use of economic modelling approaches. These models establish theoretical minimum and maximum energy consumption bounds. The minimum bound assumes that all miners employ the most energy-efficient hardware available and is calculated as follows:

$$\boldsymbol{P_{min}} = \boldsymbol{R}^{*}\boldsymbol{\eta} \tag{1}$$

here, $\boldsymbol{P_{min}}$ represents the minimum power consumption [W] (lower limit), R is the hash of the mining system [H/s], and $\boldsymbol{\eta}$ is the energy efficiency of the most efficient hardware [J/H].Condition where mining revenue equals electricity costs. Miners will cease operation if running expenses are exceeding earnings. Consequently, this upper bound can be calculated as follows:

$$\boldsymbol{E_{max}} = \frac{(\boldsymbol{I_b} + \boldsymbol{F_t})^{*}\boldsymbol{V_m}}{\boldsymbol{C_e}} \times \frac{1}{\boldsymbol{t}} \tag{2}$$

here, $\boldsymbol{E_{max}}$ represents the maximum economically viable power consumption (upper bound) [W]. $\boldsymbol{I_b}$ denotes the incentive per mined block, corresponding to the block reward [BTC], while $\boldsymbol{F_t}$ captures the total transaction fees associated with the block [BTC].$\boldsymbol{V_m}$ is the market price of a single Bitcoin [USD/BTC], $\boldsymbol{C_e}$ refers to the cost of electricity per kilowatt-hour [USD/kWh], and $\boldsymbol{t}$ specifies the time interval in hours (Fig. 5).

Key challenges and barriers to cryptocurrency adoption

Despite its innovations, the widespread adoption of cryptocurrencies faces significant hurdles [41]. These challenges can be divided into technical and operational issues intrinsic to the technology, as well as economic and regulatory complexities associated with integrating it into existing societal frameworks. Table 4 summarizes these challenges and their main characteristics.

While cryptocurrencies and blockchain technologies offer new possibilities for data and transaction management, their high energy consumption remains a major obstacle to growth. At the same time, the modern power system contends with several pressing issues. The increasing share of renewable energy sources, the need for efficient storage solutions, and the demand for reliable ancillary services have collectively made grid management more complex. Consequently, understanding the challenges and opportunities that cryptocurrencies introduce to the power sector is essential. To establish this context, the next section reviews the current state of the modern power system across three key areas.

The energy transition: technological imperatives and foundational challenges

The preceding section identified PoW mining as a major source of energy demand. Importantly, this growing load is emerging at a time when power systems themselves are undergoing a paradigm shift. To understand how these dynamics interact, it is essential to first examine the principal challenges currently being confronted by modern grids.

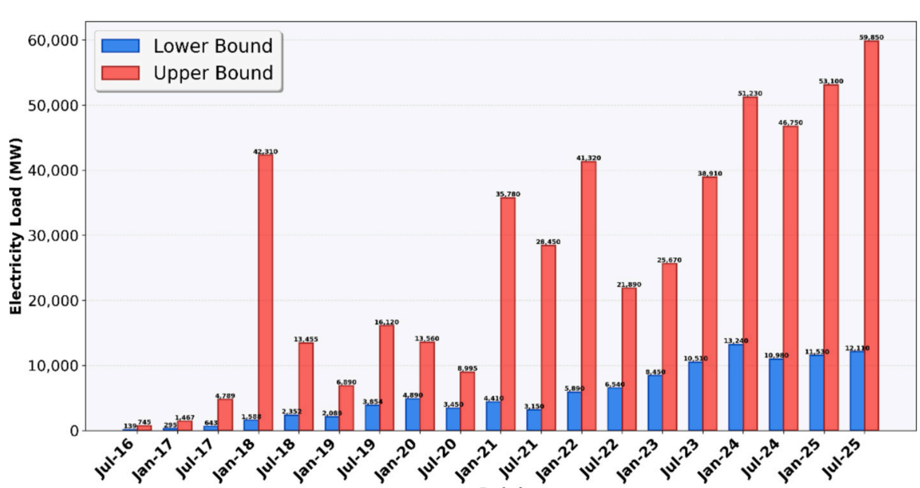

Fig. 5. Estimated upper bound (economic viability) and lower bound (hardware efficiency) of the Bitcoin network's total electricity load (2016–2025).

Table 4
A summary of key challenges to cryptocurrency adoption.

Challenge	Core description	Example/impact	Ref.
Scalability limitations	Low transaction throughput compared to traditional systems, leading to network congestion.	Bitcoin's ~7 TPS vs. Visa's thousands of TPS.	[33]
Security risks	Vulnerabilities in surrounding infrastructure (exchanges, wallets) despite a secure core protocol.	The Mt. Gox exchange hack (theft of >700 k BTC).	[21]
High energy consumption	Proof of Work algorithms require vast amounts of electricity, raising environmental concerns.	Bitcoin's annual consumption surpassing that of countries like Argentina.	[34]
High price volatility	Extreme price fluctuations due to speculation, impeding its use as a stable store of value.	Periods of >80 % value loss for Bitcoin.	[42]
Legal & regulatory uncertainty	Lack of clear, harmonized legal frameworks hinders institutional adoption and creates risk.	Ambiguity in taxation, AML/KYC laws across different nations.	[43,44]
Use in illicit activities	Semi-anonymous features are exploited for financial crime, challenging regulators.	Use in darknet markets like "Silk Road."	[45]

The challenge of fossil fuels and the shift towards sustainability

Population growth and technological advancements have driven an exponential rise in global energy consumption. Electricity generation has historically been dominated by fossil fuels because of their reliability and efficiency; however, their use is associated with severe and growing environmental consequences, as illustrated in Fig. 6. Beyond environmental concerns, fossil fuels are recognized to exert a direct economic impact since their prices continue to escalate with sustained consumption. Projections indicate that by 2050 fossil fuel prices may be expected to double, and by 2100 they could be projected to triple [46], thereby posing significant economic and sustainability challenges.

The rising cost of fossil fuels directly affects the final price of electricity generation and is recognized to pose serious challenges to energy security, particularly for nations dependent on fuel imports. According to the 2024 International Energy Agency (IEA) report, more than 75 % of global greenhouse gas emissions are generated by the energy sector, making the replacement of fossil fuels with renewable sources essential for achieving decarbonization targets [46]. Consequently, the United Nations and other international organizations have prioritized the sustainable development of renewable energy technologies [48].

Renewable sources, such as solar, wind, biomass, and hydropower, not only mitigate carbon emissions but also are recognized to enhance energy security by reducing dependence on fossil fuels, thereby highlighting the need for a fundamental transition to sustainable, low carbon energy systems. To achieve this transition, researchers are exploring three primary approaches: improving the efficiency of devices and equipment, developing advanced energy conversion technologies, and increasing the deployment of renewable energy systems that can be harnessed from natural resources with minimal environmental impact. The following section focuses on one of the most crucial of these strategies: the expansion of renewable energy adoption [49].

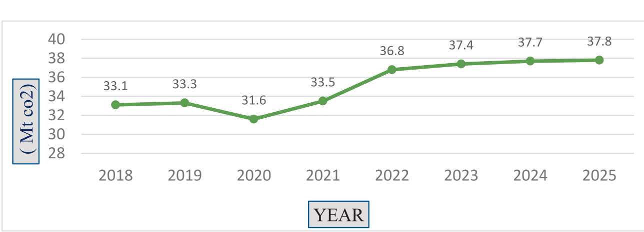

Fig. 6. Global Energy-Related CO2 Emissions (in Mt CO2), 2018–2025 [47].

A comparative analysis of renewable energy technologies and their operational limitations

Given the growing global emphasis on renewable energy, driven by policies aimed at reducing carbon emissions and ensuring sustainable, cost-effective power generation, it is essential to not only identify these energy sources and their key characteristics but also to analyze their operational challenges and explore practical solutions. This section introduces the main types of renewable energy sources, outlining their advantages and limitations. Table 5 presents a detailed comparison of renewable technologies along with their key performance attributes.

According to the 2024 International Energy Agency (IEA) report, the global adoption of renewable energy for electricity generation is observed to continue growing at a steady pace. The report's analysis highlights that the deployment of specific renewable technologies across different regions is strongly influenced by local geographic conditions, climate factors, and the availability of natural resources [61] (Fig. 7).

While renewable energy sources offer unique benefits and are expected to play a significant role in future energy supply, they present two main challenges: (1) their inherent intermittency and (2) the difficulty of matching generation with consumption in real time. These challenges can be observed to slow the development of renewable energy systems and delay progress toward a sustainable, low carbon energy future. To address these issues, researchers have proposed solutions that are considered both technically feasible and economically viable. One of the most promising approaches is recognized to be the use of energy storage systems for optimized energy management, which will be examined in the following section [62,63].

The challenge of intermittency results not only in supply shortages but also in periods of substantial energy surplus, during which wind and solar generation are observed to exceed demand, often leading to costly curtailment. Consequently, this situation has created opportunities for innovative demand-side solutions. One promising approach is recognized to be the integration of cryptocurrency mining, whose high energy intensity and geographic flexibility make it uniquely suited to act as a "buyer of last resort." By collocating mining operations with large scale renewable projects, excess electricity that would otherwise be wasted can be utilized productively. This potential synergy where cryptocurrency mining supports the economic viability of renewable energy projects will be examined in detail in Section 4.

Techno-economic assessment of conventional energy storage and the case for alternative frameworks

Although the transition to renewable energy sources is critical for achieving long-term sustainability, it also introduces considerable technical challenges for the power grid. As Denholm et al. [6] emphasize, the variable and unpredictable output of solar and wind resources can be observed to threaten grid stability, underscoring the need for additional flexibility mechanisms. Within this context, energy storage systems play a pivotal role. These systems are designed to mitigate generation variability, enhance supply reliability, and improve grid flexibility, thus forming a crucial enabler for the large-scale integration of renewable energy. Energy storage technologies are commonly classified into five categories based on the form of stored energy, as described by Dina Elafi [REF], and are summarized in Fig. 8.

There are several critical factors to consider when selecting appropriate energy storage devices. The first is capacity, which determines the total amount of energy that can be stored within the system. Equally important is the power rating, which defines the rate at which energy can be charged into or discharged from the device. Another key aspect is

Table 5
A multi-criteria comparison of leading renewable energy technologies, summarizing their techno economic characteristics and deployment status.

Renewable energy type	Technologies mentioned	Applications	Advantages	Challenges/ limitations	Major projects	Ref.
Solar energy	PV (Photovoltaics) CSP (Concentrated Solar Power) Bifacial PV Modules	Residential use, transportation, desalination, drying, irrigation	Globally available, suitable for off-grid areas, decreasing costs due to technology improvements	Intermittency, Reduced performance during Cloudy weather	MW, India)Noor CSP (580 MW, Morocco) Benban Solar Park (1600 MW, Egypt)	[46]
Wind energy	HAWT (Horizontal Axis Wind Turbines)VAWT (Vertical Axis Wind Turbines) Vortex Bladeless Turbines	Electricity generation from surface and high-altitude winds	High energy yield, new bladeless designs with lower environmental impact	Wind dependency, reduced output on calm days	Gansu Wind Farm (7965 MW, China)Hornsea Project Two (1800 MW, UK)Alta Wind Center (1550 MW, USA)	[48–51]
Biomass Energy	Biodiesel production Biochar generationDirect combustion Use in fuel cells	Electricity generation, pollutant adsorption, biofuel for engines	Diverse resource base, liquid fuel production, waste utilization	High viscosity of bio-oil, emissions from combustion	Ironbridge (740 MW, UK)Polaniec (220 MW, Poland) Alholmens Kraft (240 MW, Finland)	[48,52–56]
Hydropower	Large-scale hydroelectric damsPumped hydro storage systems	Large-scale electricity generation, storage integration	Reliable output, high storage capacity	Environmental impact, high infrastructure costs	Three Gorges Dam (22,500 MW, China) Itaipu Dam (14,000 MW, Brazil/Paraguay)	[57,58]
Green Hydrogen & Fuel Cells	Electrolysis powered by renewables Application in fuel cells	Long-term energy storage, power supply, transportation fuel	Emission-free, flexible integration in renewable systems	High production cost, infrastructure requirements	No specific commercial project cited; discussed in terms of development potential	[48,59,60]

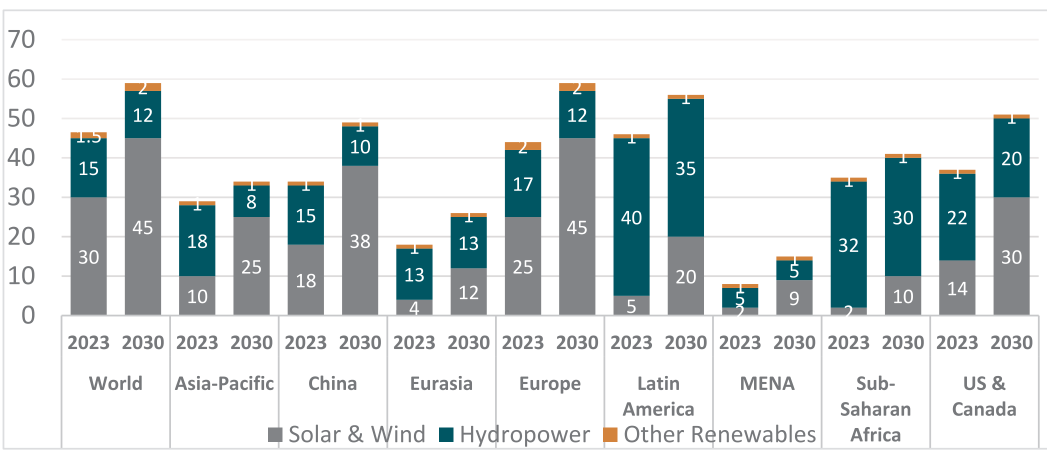

Fig. 7. Share of renewable energy by technology and region in **2023**, with projections for **2030.**

efficiency, referring to the proportion of energy lost during storage and across charge–discharge cycles. The storage duration indicates how long the energy can be retained before use, while the charge and discharge times describe the time required for the device to complete these processes. Table 6 provides a comparative overview of the aforementioned storage technologies, highlighting their characteristics from multiple technical and economic perspectives.

Among the available energy storage solutions, Electrochemical Energy Storage Systems (ECESS) are observed to stand out due to their high energy density and widespread application in electric vehicles. Lithium-ion batteries, in particular, are recognized to offer high efficiency, scalability, and rapid response capabilities, making them the leading technology within this category. Mechanical Energy Storage Systems (MESS) such as pumped hydro and compressed air energy storage are also considered a promising grid scale option, providing long operational lifetimes and compatibility with renewable generation facilities.

In contrast, Electromagnetic Energy Storage Systems (EESS) including supercapacitors and superconducting magnetic energy storage are designed to provide fast response times and exceptional cycle life but are constrained by high costs and limited scalability. Thermal Energy Storage Systems (TESS) are employed less frequently due to lower efficiency and complex system designs. Finally, Chemical Energy Storage Systems (ChESS) such as hydrogen- and ammonia based storage are expected to offer theoretically unlimited capacity but face significant infrastructure costs and technological maturity barriers.

Despite substantial progress, widespread adoption of ECESS and MESS remains hindered by several technical and economic constraints. As highlighted by Koohi Fayegh and Rosen [66], batteries are still observed to suffer from high upfront costs, limited service lifetimes, and complex thermal management requirements, while MESS projects are often required to involve costly infrastructure and are constrained by geographic location. Similar observations are reported by Elalfy et al. [46], emphasizing the urgent need for complementary and innovative storage frameworks. Consequently, despite significant advancements in conventional energy storage, their limitations prevent full exploitation of their potential. Bridging these gaps requires the development of new approaches that are designed to be both technically robust and economically feasible.

Synthesized case study: 100 MW renewable-powered mining with hybrid storage

This section bridges the summarized features of storage technologies

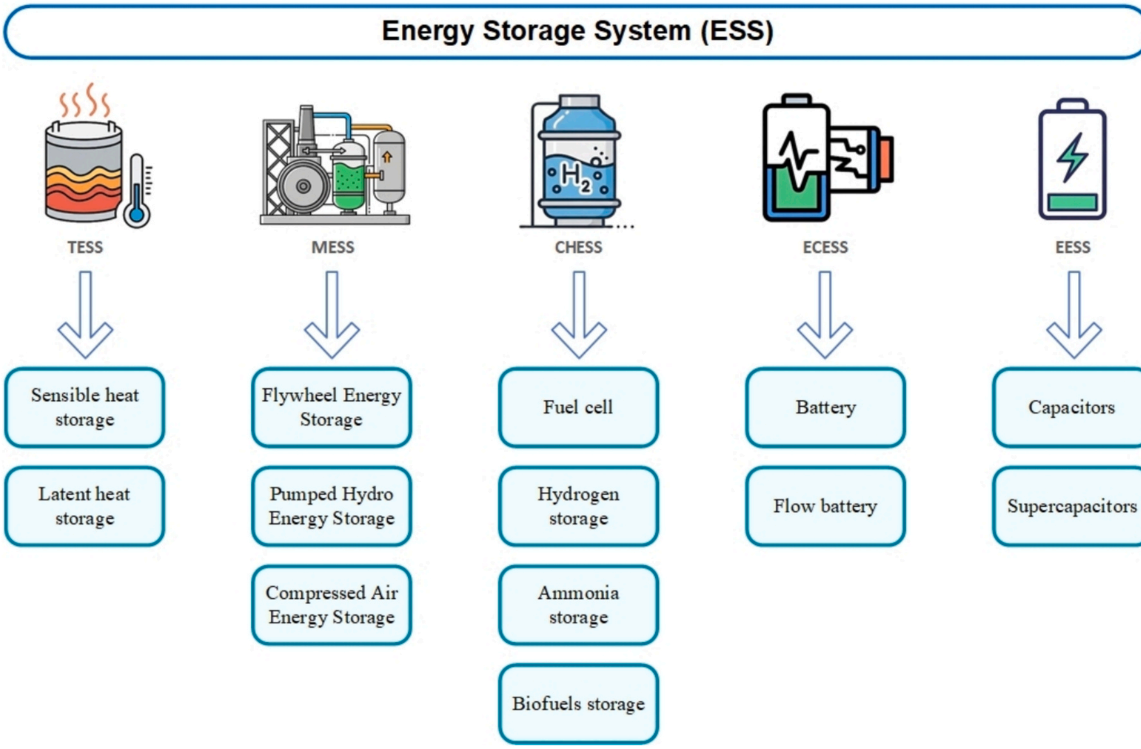

Fig. 8. Classification of Energy Storage Systems (ESS) based on the form of stored energy [46].

Table 6
A techno economic comparison of various energy storage technologies.

Energy storage type	Advantages	Disadvantages	Applications	Storage capacity (MW)	Efficiency (%)	Cost ($/kWh)	Ref
Electrical energy storage (EESS)	Fast response, flexible power output, long lifespan	High cost for large-scale storage	Power grids, electric vehicles, voltage stabilization	**0.05–10**	**85–95**	**897–8974**	[46,64,65]
Electrochemical energy storage (ECESS)	High energy density, scalability, application in electric vehicles	High cost, environmental impact, thermal management requirements	Electric vehicles, home energy storage systems, grid support	**10–13,000 KW/kg**	**60–86**	**100–1500**	[66–68]
Mechanical energy storage (MESS)	Long lifespan, high storage capacity, fast response time	High initial cost, geographical constraints	Grid stabilization, renewable power plants	**1–1000+**	**70–90**	**150–2000**	[69,70]
Thermal energy storage (TESS)	Low initial cost, reduced energy consumption, lower greenhouse gas emissions	Low overall efficiency (30–50 %), thermal losses, complexity	Heating systems, industrial heat recovery, solar energy storage	Variable (depending on technology)	**50–99**	**100–800**	[71,72]
Chemical energy storage (ChESS)	Long-term storage capability, application in power generation and transportation	Low efficiency, high infrastructure costs, some methods still under research	Power generation, transportation, industry	Unlimited (depending on scale)	**40–60**	Low (depending on technology)	[73–77]

Table 7
Quantitative lifecycle comparison of storage frameworks for a 100 MW curtailment-focused scenario.

Metric	BESS (Li-ion)	PHS (pumped hydro)	VESS (crypto mining)	Ref.
CAPEX (USD/kW)	1,200–1,800 (Declining with scale)	1,999–5,505 (High, site-dependent)	≈0 (Cost tied to mining hardware)	[78,79]
LCOS (USD/MWh)	115–254 (4 h systems)	150–225 (8–12 h systems)	Market-dependent (Driven by crypto–power price gap)	[78–80]
Lifespan (Years)	15–20	60–80+	1.5–3 (ASIC replacement)	[78,81,82]
Round-trip efficiency	85–90 %	70–85 %	N/A (Unidirectional)	[78]
Primary use case	Fast-response/grid services	Long-duration/seasonal storage	Renewable curtailment absorption	[82]
Carbon footprint (g CO_2 e/kWh)	116–250 (High embodied carbon)	Lowest among mature tech	Zero operational (if 100 % renewable)	[82,83]
Main environmental concern	Lithium/cobalt extraction; end-of-life toxicity	Land & water use; construction impact	Rapid e-waste generation	[81,82]

(Table 6) with their real-world function in cryptocurrency mining. The case study focuses on a 100 MW mining facility built alongside a 150 MW hybrid solar–wind farm located in a region with strong renewable potential but limited transmission lines areas such as West Texas in the U.S. or Western China. These regions often waste part of their renewable generation, which makes them ideal testbeds for assessing different storage and flexibility solutions aimed at putting surplus power to use. Following the storage typologies in Table 6, a comparative life-cycle and carbon footprint analysis was performed on three representative storage configurations. Table 7 highlights the main numerical findings from this comparison.

(a) Electrochemical Storage (Li-ion BESS): Li-ion battery systems represent the most widespread form of electrochemical energy storage (ECESS) in utility-scale applications. In this analysis, a 100 MW/400 MWh (4-hour duration) system is considered.

- **Lifecycle Economic Perspective:** BESS offers high scalability and round-trip efficiency (85–90 %), but as of 2024–2025, it remains capital-intensive, with CAPEX estimated at 1,200–1,800 USD/kW [84]. Despite relatively low OPEX, its (15– 20) year lifespan is limited by chemical degradation. The Levelized Cost of Storage (LCOS), ranging from 115 to 254 USD/MWh [78], indicates that BESS is more suited to fast-response ancillary services (e.g., frequency regulation [82]) than to large-scale, multi-hour energy shifting.
- **Carbon Footprint Perspective:** The main environmental burden of BESS arises from the embodied carbon in material extraction and cell manufacturing. This "carbon debt," estimated at 116–250 g CO_2 e/kWh [82,83], is gradually offset during its operational lifetime through the displacement of fossil-based generation.

(b) Mechanical Storage (Pumped Hydro – PHS): Pumped hydro storage (PHS) is the most mature and widely deployed mechanical energy storage (MESS) technology, typically offering capacities exceeding 1 GW.

- **Lifecycle Economic Perspective:** PHS deployment is constrained by geography, requiring suitable elevation and water resources. Capital costs are relatively high (1,999–5,505 USD/kW [79]), but the long operational lifespan (60–80 + years) significantly reduces lifecycle costs. The LCOS is estimated at 150–225 USD/MWh for 8–12-hour duration systems [78,79], with round-trip efficiency between 70–85 %.
- **Carbon Footprint Perspective:** Although the construction phase entails substantial emissions due to large-scale use of concrete and steel, the long service life of PHS results in low lifecycle carbon intensity per delivered MWh, positioning it among the most sustainable large-scale storage options [78,83].

(c) Virtual/Economic Storage (Cryptocurrency Mining as VESS): When operated flexibly, cryptocurrency mining can function as a Virtual Energy Storage System (VESS) an "economic battery" that converts surplus renewable energy into digital assets.

Lifecycle Economic Perspective: In this framework, the CAPEX linked to storage functionality is essentially zero, because the mining hardware (ASICs) represents the primary business investment rather than a grid asset. Energy is consumed unidirectionally, with no round-trip losses. Consequently, this configuration enables multi-hour absorption of renewable generation that would otherwise be uneconomic for electrochemical storage. The main economic risk, however, arises from market volatility, specifically the fluctuating value of the generated asset (see Section 4.1.1).

- **Carbon Footprint Perspective:** When operated entirely on curtailed renewable energy, operational carbon emissions are essentially zero. However, the lifecycle footprint is influenced by the high rate of e-waste generation, as ASIC devices typically have a functional lifespan of 1.5–3 years, leading to continuous manufacturing and disposal cycles [81].

Ancillary services: a vital component and economic challenge in modern power systems

The operational landscape of modern power systems is observed to be undergoing a profound transformation. Driven by global decarbonization efforts, conventional fossil fuel synchronous generators are progressively being replaced by inverter-based resources. While this shift is designed to advance environmental objectives, it is also associated with significant and unprecedented challenges for maintaining grid stability and reliability [85]. The resulting reduction in system rotational inertia, together with the variable and uncertain output of renewable energy sources, has been shown to markedly increase the need for redesigning and reinforcing grid control strategies. Within this evolving context, ancillary services are recognized to have evolved from a secondary function into a core operational requirement, playing a critical and dynamic role in ensuring real-time reliability and the stable operation of contemporary power grids [86] (Fig. 9).

Architecture and classification of ancillary services in high IBR power networks

The modern architecture of ancillary services addresses physical challenges that vary across different time scales. Because system inertia is decreasing, the Rate of Change of Frequency (RoCoF) rises sharply after disturbances, creating a need for ultra-fast response mechanisms.

Consequently, services such as Fast Frequency Response (FFR) and virtual inertia, delivered by inverter based resources (IBRs) using grid forming control, are essential for maintaining frequency stability in this low inertia environment [87]. A comprehensive classification of these key ancillary services, arranged according to their temporal hierarchy and technical objectives, is presented in Table 8.

The economic dynamics of ancillary services: cost trends and market challenges

The growing importance of ancillary services has been accompanied by a significant surge in their associated economic costs. In major system operator markets, the total value of ancillary services now amounts to several billion dollars annually. For instance, in 2023, the total ancillary services market expenditures in the PJM Interconnection in the United States exceeded 3 billion, highlighting the considerable financial burden associated with these services [91].

Historical data from electricity markets clearly illustrates a sustained upward trend in ancillary service costs. The roots of this price escalation are complex, but market analysts, such as Müller et al. [92] in their study of the German market, attribute it to two key factors: (1) the retirement of traditional thermal plants that were low cost providers of these services, and (2) the increasing need for fast and flexible reserves to manage the inherent uncertainty of inverter based resources (IBRs), a challenge modelled by Fang et al. [88].

As a concrete illustration, the average capacity price for secondary reserves (aFRR) in the German market increased from about 5/MW/h in 2018 to more than 20/MW/h by early 2023, marking a rise of over 300 % in less than five years [92]. Likewise, in the United Kingdom, the significant economic value of rapid-response services in low inertia power systems has been demonstrated through the introduction of new ultra-fast mechanisms, such as Dynamic Containment, where average capacity prices have ranged between 17 and 35 per megawatt-hour [89].

As a concrete illustration, it can be observed that the average capacity price for secondary reserves (aFRR) in the German market was raised from approximately 5 MW.h in 2018 to over 20 MW.h by early 2023, representing more than a 300 % increase within less than five years [92]. Similarly, in the United Kingdom, the substantial economic value of rapid response capabilities in low inertia power systems has

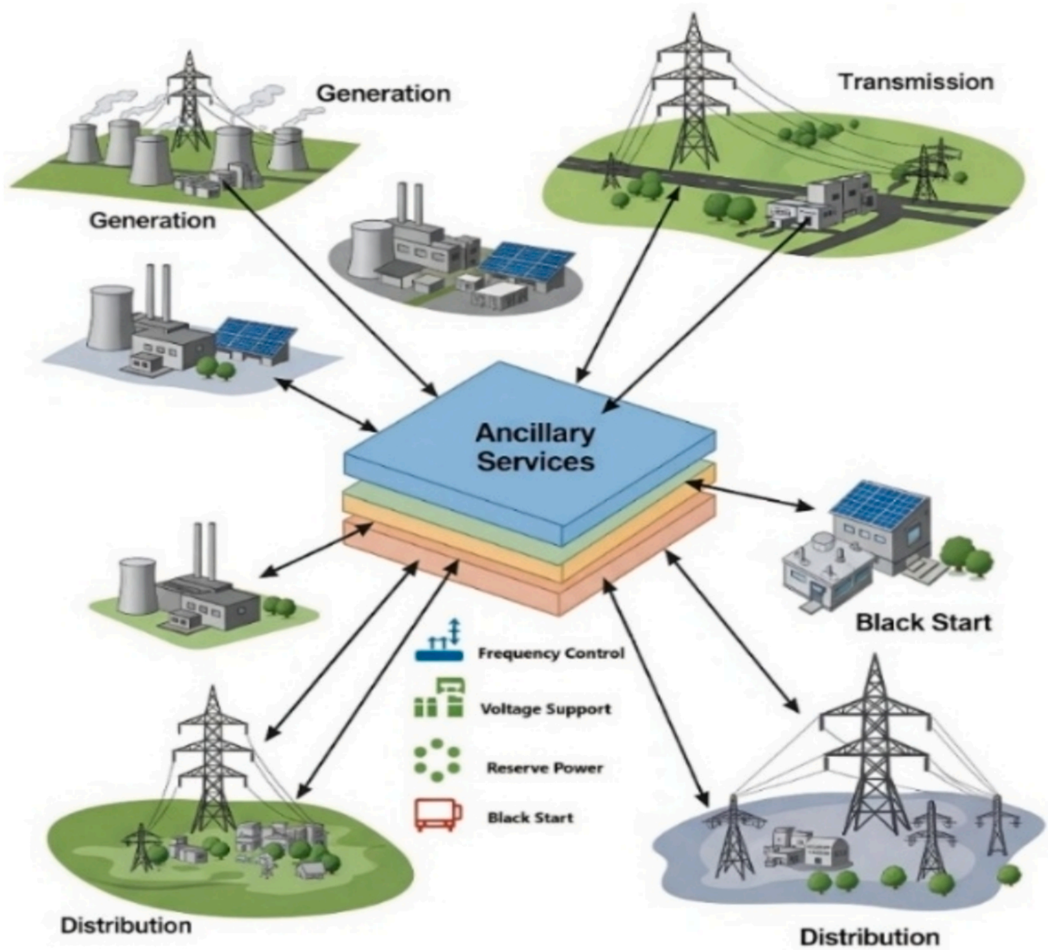

Fig. 9. The architecture of ancillary services in a modern power system.

Table 8
A hierarchical classification of modern ancillary services based on their technical objectives and response time scales [88–90].

Service category	Service name (abbreviation)	Main technical objective	Response time scale	Key providers in the modern paradigm
Instantaneous response	Virtual Inertia/Fast Frequency Response (FFR)	To mitigate the Rate of Change of Frequency (RoCoF) and frequency nadir, thereby enhancing system stability following sudden disturbances.	**100 ms to 2 s**	Battery Energy Storage Systems (BESS), Supercapacitors, Grid-Forming Inverters, very fast-acting responsive loads.
Primary response	Frequency Containment Reserve (FCR)	To stabilize the system frequency at a new equilibrium point post-disturbance, preventing critical frequency oscillations.	**2 s to 30 s**	Synchronous Generators, BESS, wind and solar power plants with advanced controls, aggregators of Distributed Energy Resources (DERs).
Secondary response	automatic Frequency Restoration Reserve (aFRR)	To restore the frequency to its nominal value and maintain the power balance between control areas, ensuring stable system operation.	**30 s to 15 min**	Highly flexible generation units, BESS, Vehicle-to-Grid (V2G) capable electric vehicle fleets, controllable industrial loads.
Tertiary response	manual Frequency Restoration Reserve (mFRR)	To economically optimize power system operation and manage grid congestion using long-duration storage and generation units with low start-up costs.	**>15 min**	Generation units with low start-up costs, long-duration energy storage, and flexible load management.
Voltage control	Reactive Power Support	To maintain the voltage profile within permissible static and dynamic limits, ensuring power quality and enhancing grid stability.	Instantaneous to continuous	Static Synchronous Compensators (STATCOM), Static VAR Compensators (SVC), Smart Inverters, Synchronous Generators.

been highlighted by the deployment of new ultra-fast services, such as Dynamic Containment, with average capacity prices reported to range from 17 to 35 per megawatt hour [89].

As shown in Table 9, the cost of each ancillary service is strongly influenced by its technical characteristics. The inherent uncertainty associated with inverter based resources (IBRs) directly affects these costs. Studies indicate that a 10 % increase in the penetration of variable renewable energy sources can result in a 15–20 % increase in the need for operational reserves, which in turn drives up clearing prices in ancillary service markets [88].

To provide a precise, data driven assessment of these costs, the ancillary service market of the PJM Interconnection in the United States is examined as a case study. Being the largest electricity market in North America, PJM is regarded as an appropriate benchmark due to its data transparency and advanced market structure [91]. Within this market, services are defined by specific products, such as Regulation (RegA/RegD) and Synchronized Reserve, with compensation being determined through a pay for performance framework.

Table 9
Economic analysis of key ancillary services in the PJM market (based on 2023 data) [93].

Ancillary service type	Capacity cost (availability)	Utilization cost (performance/ energy)	Key cost drivers
Fast Frequency Response (Regulation Signal D– RegD)	10–25 per MW-hour (Capability Payment)	8–15 per ΔMW (Performance “Mileage” Payment)	High capital expenditure (CAPEX) for batteries and inverters; requirement for rapid and precise response.
Primary Reserve (Synchronized Reserve)	2–10 per MW-hour(Avg. Annual Price: ~6.23)	Paid at Locational Marginal Price (LMP)	Opportunity cost for spinning units; fuel costs to maintain minimum generation levels.
Secondary Reserve (Regulation Signal A – RegA)	8–20 per MW-hour (Capability Payment)	4–10 per ΔMW (Performance “Mileage” Payment)	Variable costs (fuel and wear-and-tear) from continuous ramping of thermal generators.
Tertiary Reserve (Non-Synchronized Reserve)	0.5–2 per MW-hour (Avg. Annual Price: ~1.02)	Paid at LMP + Start-up Costs	Low availability cost (offline units) but high start-up and ramping costs upon dispatch.

Furthermore, Table 9 presents a comparative economic analysis of key ancillary services within the PJM market, based on official 2023 data. It highlights the distinct capacity (availability) and utilization (performance) costs associated with each service, providing a clear picture of the economic dynamics that are observed in modern ancillary service markets [90].

3.3.3. Emerging solutions: enabling digital flexibility on the demand side

Rising costs and technical constraints associated with procuring ancillary services from traditional sources have increasingly strengthened the economic case for leveraging alternative solutions, particularly on the demand side. Demand Response (DR) has been identified as a highly promising and cost effective option. According to recent projections by the U.S. National Renewable Energy Laboratory (NREL), up to 200 GW of demand side flexibility could be contributed by 2030, representing nearly 20 % of peak load. Consequently, this shift is expected to offer significant potential for reducing overall system costs [94].

Among emerging demand side resources, computational loads, including data centres and cryptocurrency mining operations, are recognized for their distinctive operational characteristics. These loads are inherently fast, precise, and digitally controllable, thereby enabling high quality ancillary services to be delivered. Unlike conventional industrial consumers, cryptocurrency mining facilities can have their power consumption reduced by large margins in under one second, as demonstrated by Gao et al. [95] in their analysis of digitally controlled loads. This ultra-fast response capability renders them particularly well suited for high value ancillary services, including Fast Frequency Response (FFR) and Frequency Containment Reserve (FCR).

Economic analyses further highlight this potential. For instance, it was found in a study conducted in the Texas ERCOT market that a 100 MW Bitcoin mining facility, when being enrolled in the Responsive Reserve Service, could earn several million dollars annually while simultaneously contributing to grid stability in a system characterized by high wind energy penetration [96]. This strategic alignment of economic viability and technical value is underscored by the integration of flexible computational loads into ancillary service market designs, thereby enabling a clean, reliable, and cost effective energy transition.

Additionally, the significant technical and economic pressures faced by the modern grid are emphasized by the integration of intermittent renewable energy sources and the rising costs of ancillary services. In light of this, the clear need for novel and cost effective flexibility solutions is recognized, and the following section will examine how the unique operational characteristics of cryptocurrency mining can be leveraged strategically, transforming it from a system burden into a valuable grid asset.

Challenges and opportunities of cryptocurrency mining in power systems

The rapid proliferation of blockchain based technologies, particularly cryptocurrencies such as Bitcoin, has led to the emergence of a new category of industrial load on power grids. This load is recognized not only for its substantial energy consumption but also for its operational patterns, which can be either conflicted with or complemented by the dynamic behaviour of the grid [90]. Based on the PoW algorithm, cryptocurrency mining is characterized by massive electricity requirements, positioning it as one of the most critical topics at the intersection of energy policy, power systems engineering, and environmental sustainability in recent years [97]. On one hand, significant challenges are presented by cryptocurrency mining, including increased base load, demand fluctuations, elevated carbon emissions, and additional stress imposed on generation and transmission infrastructure. On the other hand, its potential as a flexible load or even as a “virtual battery” has been emphasized by researchers and industry experts [98]. In this regard, mining operations can be strategically activated during periods of excess energy particularly from renewable sources and curtailed rapidly during peak demand or supply shortages. Consequently, this inherent flexibility positions cryptocurrency mining as a promising tool for smart demand-side management and grid frequency regulation [99].

This section provides a comprehensive literature review of the environmental and grid level impacts associated with cryptocurrency mining. Global experiences and case studies from countries pursuing crypto mining for economic, technological, or regulatory objectives are examined. Additionally, studies exploring the feasibility of mining farms being utilized as interruptible loads, price responsive demand resources, or active participants in electricity markets are incorporated.

The primary objective of this section is to establish a structured framework through which the current landscape can be critically analyzed and the potential role of cryptocurrency mining in the architecture of modern power systems can be assessed. Within this framework, the challenges and opportunities posed by cryptocurrency mining are analyzed across multiple dimensions and scales. These challenges and opportunities are categorized into four principal areas, with the analysis systematically conducted from each perspective.

Cryptocurrency mining as a virtual energy storage system: technical framework and economic potential.

As discussed in Section 3.2, conventional physical energy storage systems (EESS), such as batteries, are considered critical components in modern power systems; however, they continue to face notable challenges. These challenges include high capital costs, energy losses due to round trip inefficiency, and constraints imposed by geographical and environmental factors [99]. In response to these limitations, the innovative concept of a Virtual Energy Storage System (VESS) has been introduced [100].

Within this context, a specialized VESS approach utilizing cryptocurrency mining has been proposed, referred to as the Cryptocurrency Energy Storage System (CESS). The fundamental principle, as formulated by Hajiaghapour-Moghimi et al. [101], involves converting surplus renewable energy rather than allowing it to be curtailed or stored in costly batteries into and storing it as cryptocurrency units (CCUs), such as Bitcoin (BTC). In this framework, the system is “charged” during periods of low demand through energy consumption for mining and "discharged" during peak periods by leveraging the stored economic value to offset microgrid operational costs. Consequently, the concept of "efficiency" is shifted from a purely technical metric to an economic one, providing advantages such as minimal maintenance costs, effectively unlimited storage in digital wallets, zero energy related capital investment, and an indefinite lifecycle. (Fig. 10).

The potential of this approach has been examined from multiple perspectives. Based on real world data from a Finnish microgrid, Hajiaghapour Moghimi et al. [101] demonstrated that the integration of CESS in a grid connected mode could reduce operational costs by up to 46.5 % while nearly eliminating renewable energy curtailment. A high Return on Investment (ROI) of 44.2 % in islanded mode and 51.6 % in grid connected mode was reported, highlighting the economic viability of the method. In a subsequent study [102], the same research group developed a control and hedging mechanism for optimally selecting between "exporting electricity to the grid" and "mining cryptocurrency" based on real-time profitability. Simulations for a residential building in Helsinki revealed a 68.1 % reduction in annual costs and an ROI of 57.7 %, significantly enhancing incentives for solar energy investment.

While the studies discussed above primarily focus on CESS as a tool for operational optimization, other research has investigated its potential to enhance overall project economics. Bitcoin mining was proposed by Xie et al. [103] as a prime example of "Productive Use of Energy" (PUE) and as an "anchor load" for solar microgrids in developing regions. Their analysis demonstrated that cryptocurrency mining can function as a flexible "Economic Battery," storing energy not in electrochemical form but as a digital asset with global value. As a result, a stable revenue stream is created that is independent of local electricity demand, significantly reducing the payback period and increasing the investment rate of return.

The concept of VESS has been extended beyond cryptocurrency applications. For instance, in a related study [104], Xie et al. examined the use of controllable loads, such as air conditioning systems, as VESS. Their findings revealed that participation in VESS through demand side management reduces the required investment in physical battery energy storage systems (BESS), and that the optimal BESS size is more sensitive to load variability than to fluctuations in solar generation. Collectively, these results indicate that flexible loads whether in the form of cryptocurrency mining or other controllable consumption are essential for enhancing the economic viability and sustainability of future microgrids.

The analysis demonstrates that the CESS concept is a highly promising economic solution for managing surplus energy while promoting investment in renewable resources. Unlike conventional thermal storage, long term energy retention is enabled in the form of a financial asset. Nevertheless, despite the notable economic benefits, the technical implications for the grid cannot be overlooked. Of particular importance is the impact of these large, flexible, and non-linear loads on grid performance, especially regarding ancillary services. Addressing this issue constitutes the central focus of the following section.

Economic viability and the dual-volatility operational framework

A central topic in the literature on Virtual Energy Storage Systems (VESS) concerns their economic behaviour under the influence of cryptocurrency price volatility [80]. Studies indicate that the economic performance of VESS depends critically on how the system responds to simultaneous fluctuations in two key markets: electricity and cryptocurrency. To analyze this, the literature generally examines two interconnected but distinct optimization challenges: operational optimization, focusing on mining decisions, and financial optimization, relating to the timing of asset liquidation.

VESS operations are often described as a dual-volatility system, in which both electricity prices and cryptocurrency values fluctuate simultaneously [80]. This interaction generates a range of operational states, commonly represented as a four-quadrant matrix of electricity and crypto price combinations. Low electricity/high crypto prices correspond to ideal conditions for profit maximization. High electricity/high crypto prices are characterized by the opportunity cost of curtailment [80], where the grid must provide incentives sufficient to offset potential mining profits. High electricity/low crypto prices describe conditions that can lead to operational curtailment or shutdowns, particularly during prolonged bearish market periods. Low electricity/low crypto prices correspond to marginal or breakeven operation. Quantitative modelling approaches such as Stochastic Optimization and Real Options Analysis (ROA) are used to estimate optimal operational thresholds under volatile market conditions [80]. These frameworks are primarily applied as risk-hedging strategies to ensure operational resilience.

In addition to operational considerations, the literature highlights

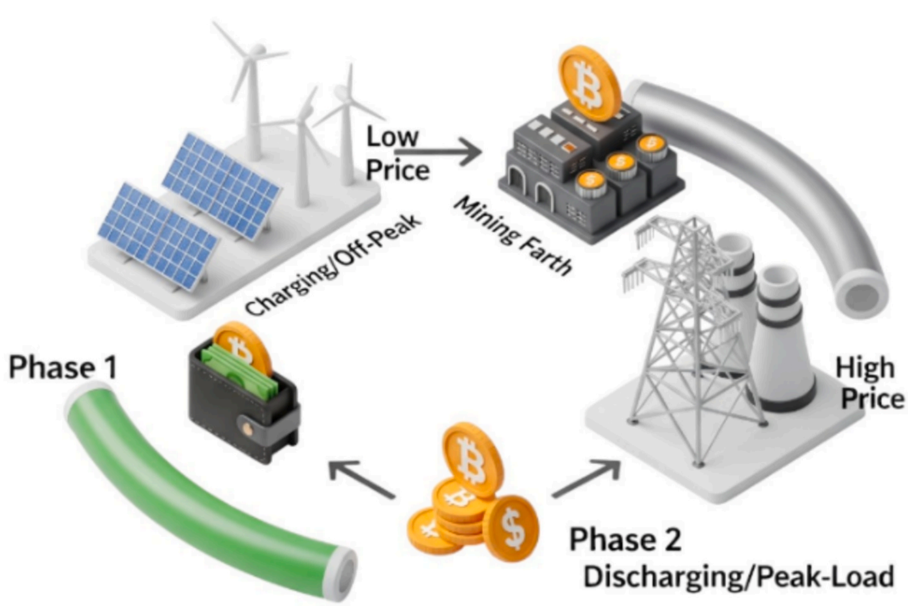

Fig. 10. The cryptocurrency energy storage system (CESS) conceptualized as an "economic battery," redefining energy storage as the conversion of electricity into a tradable digital asset.

that the act of selling mined digital assets is decoupled from mining itself. Unlike physical batteries that must discharge before recharging, VESS allows mined assets to be stored indefinitely in a digital treasury, introducing a secondary optimization challenge: determining the optimal timing for asset liquidation to maximize cumulative returns. Research in this area generally focuses on two themes. First, price prediction feasibility: systematic reviews such as Sebastião & Godinho (2021) [105] demonstrate that cryptocurrency price forecasting is a mature field. Machine Learning and Deep Learning models, including LSTM and GRU, show strong performance in short-term trend prediction and volatility estimation, forming a robust methodological basis for strategic decision-making. Second, optimal liquidation and portfolio management: recent studies have developed dynamic, data-driven strategies for asset management. For instance, Stavroyiannis & Zarkos (2023) [106] propose Deep Reinforcement Learning (RL) models in which an AI agent learns an optimal policy for gradual asset liquidation, effectively managing the digital treasury to maximize cumulative profit.

The reviewed studies collectively indicate that cryptocurrency price volatility is not a structural limitation of VESS but a manageable factor within operational and financial decision-making. Two complementary mechanisms emerge in the literature: operational risk management using quantitative models such as Real Options Analysis to guide mining decisions [80], and financial optimization through AI-based forecasting [105] and reinforcement learning strategies [106] to determine asset sale timing. This integrated perspective, drawing from energy economics, computational finance, and data science, emphasizes that the economic viability of VESS depends on adaptive, data-driven decision frameworks rather than static cost–benefit evaluations.

4.2. Integration of cryptocurrency mining into ancillary service provision in modern power grids

As discussed in Section 3.3, the emergence of cryptocurrency mining as a large energy consumer has raised significant concerns regarding power grid stability. Nevertheless, the distinctive characteristics of these loads particularly their high flexibility, rapid controllability, and location independence have attracted research attention to their potential for providing ancillary services and supporting grid management. Consequently, mining farms can be transformed from potential threats into valuable instruments for enhancing power system stability and operational efficiency. The existing literature in this domain can be categorized into three primary groups, depending on the grid level at which impacts and services are evaluated: the generation level, the transmission level, and the distribution level. In this section, an analytical review of key studies conducted across these areas is presented.

Pivotal contributions of cryptocurrency mining to generation-scale stability enhancement and flexibility provision.

At the generation level, large scale mining centers are increasingly regarded as Controllable Load Resources (CLRs). Such resources can assist system operators in maintaining the real time balance between generation and consumption, thereby enhancing grid stability and supporting the transition to clean energy. As shown in Table 10, research at this level ranges from complex optimization models for market participation strategies [107] to dynamic simulations illustrating physical capabilities [108]. A common methodology employed in these studies involves the use of real electricity market data, particularly from ERCOT, for model validation, emphasizing the maturity of this research field.

The primary innovation highlighted in these studies lies in the

Table 10
A critical review of recent studies on the techno-economic feasibility of miners providing ancillary services.

Ref	Main research objective	Type of ancillary Service	Methodology/model	Tools and data	Key findings	Main contribution	Limitations and technical remarks
[107]	Optimizing miner participation strategies in ancillary service markets to maximize profit	Frequency regulation (Reg-Up/Down) and general ancillary services	Mathematical optimization (offline and online); greedy algorithm for offline, stochastic gradient descent for online	Real data from ERCOT ancillary services market and Bitcoin prices;; 250 MW mining center with heterogeneous miners	Proposed algorithm increased miner profit by over 20 %; optimal strategy involves dynamic hourly allocation, not fixed capacity commitment	Introduced a mathematical formulation for optimizing participation with heterogeneous devices; developed both offline and online risk-aware models	Did not consider communication delays in control signals; assumed miners are price-takers; strategic competitor behavior not modeled
[108]	Techno-economic feasibility assessment of providing frequency regulation by mining centers	Frequency regulation (Reg-Up/Down)	Economic modeling for decision-making; transient dynamic simulation to assess physical effects	Real ERCOT market data **(2019–2022);** simulated **2000**-bus Texas power system model	Miners exhibit high ramp rates (**50 MW/s**); profitable during grid contingency events	Demonstrated physical feasibility and fast response of miners via transient simulation and economic analysis	Results are highly dependent on ERCOT market design; only one ancillary service is analyzed; generalization to other markets may be limited
[109]	Developing an enhanced demand response mechanism to suppress transients from generation ramping	Ramping support	Extended unit commitment problem; solved using Pontryagin's minimum principle	Real data from Israeli grid operator, historical data from California, actual miner specifications	Optimal miner scheduling eliminated ramping costs; profitability depends on hardware costs and renewable penetration level	Extended previous models by integrating ramping costs and constraints into optimization for high-renewable networks	Assumes centralized control and full knowledge of load profiles, which is impractical; linearized profitability model
[110]	Estimating the monetary value of air pollution and health damages caused by cryptocurrency mining	(Related topic) Assessment of energy-related externalities	Air pollution modeling, exposure–response functions, economic valuation	Energy consumption and emissions data from U.S. and China; economic factors from EPA and other sources	In **2018**, each **1** of Bitcoin value caused **0.49** in health and climate damages; externalities may exceed coin value	Provided monetary estimates of negative externalities from crypto mining	Assumes national average electricity mix; exact miner locations are unknown, which affects damage distribution

recognition of miners not merely as passive loads but as active service providers with unique operational characteristics. For example, Menati et al. [107] focus on profit optimization, Rayan El Helou et al. [108] demonstrate physical capability for fast response, and Ginzburg Ganz et al. [109] present a novel application aimed at mitigating ramping issues. Moreover, adopting a broader perspective, Andrew L.Goodkind [110] outlines the role of cryptocurrency mining in the future energy ecosystem and its synergy with other clean technologies, such as green hydrogen.

Transmission-level deployment of cryptocurrency mining as a strategic resource for congestion mitigation and grid performance optimization.

The transmission level functions as an intermediary between bulk generation and distribution centers. The primary challenge encountered at this level involves managing power flows and preventing congestion in high-voltage lines. Although substantial research has been conducted on ancillary services at the generation level, such as frequency regulation, and on power quality issues at the distribution level, examining impacts at the transmission level is particularly crucial due to its broader economic implications.

In a comprehensive study, Menati et al. [111] investigated the effects of connecting large scale mining farms to the transmission grid through a detailed simulation of the Texas grid, modelled as a 2000 bus system. The analysis incorporated complex operational problems, including Security Constrained Unit Commitment (SCUC) and Security Constrained Economic Dispatch (SCED). It was found that both the location and the size of mining loads exert a decisive influence on grid performance. Improperly coordinated placement of mining farms can easily generate or exacerbate congestion on transmission lines. The resulting economic consequence of such congestion is a significant increase in the Locational Marginal Price (LMP) in affected regions, which translates into higher electricity costs for all consumers within these areas. (Fig. 11).

The primary innovation of the study is found in presenting an endogenous solution to this challenge. It is demonstrated that the participation of miners in demand response programs, particularly price based schemes, creates a mutually beneficial strategy. On one hand, excessive costs can be avoided and additional revenue generated by miners through the reduction of their consumption during peak congestion periods. On the other hand, the grid benefits from mitigated line loading, enhanced reliability, and moderated electricity prices. This research also highlights a notable gap. Although short-term operational impacts have been extensively studied, further investigation is required regarding the optimal siting of mining farms and their long-term implications for Transmission Expansion Planning (TEP).

Distribution network impacts of cryptocurrency mining: advancing power quality and prolonging infrastructure lifespan.

At the distribution level, the research focus is shifted from bulk system services to localized challenges. The effects of connecting cryptocurrency miners on power quality, equipment loading, and the safety of low voltage networks are assessed. These insights are considered particularly important for Distribution System Operators (DSOs), who are tasked with maintaining the stability and reliability of local grids. Table 11 presents a detailed review and analysis of the existing studies in this area.

Research at the distribution level is approached from a dual perspective regarding the impact of cryptocurrency miners. On one hand, due to the use of switching mode power supplies with Power Factor Correction (PFC) circuits, a near unity or even leading power factor is exhibited, which contributes to enhancing the local grid's voltage profile. This is recognized as a notable advantage for distribution networks.

On the other hand, the inherently non-linear characteristics of these loads result in the injection of substantial harmonic currents into the grid, which may potentially exceed the limits defined in the IEEE 519 standard. The most critical concern at this level, however, is represented by the effect of miners' constant, high consumption load profiles on grid infrastructure. Empirical evidence indicates that the uncontrolled penetration of residential mining operations can lead to a significant increase in coincident loads, accelerate the aging of distribution transformers, and thereby reduce their operational lifespan.

These findings emphasize that the primary challenge is not posed merely by the presence of these loads but by their uncoordinated deployment. Consequently, solutions are recommended to focus on the development of active management and control frameworks by Distribution System Operators (DSOs), similar to the theoretical framework proposed by Karimi et al. [116] for optimal management of flexible

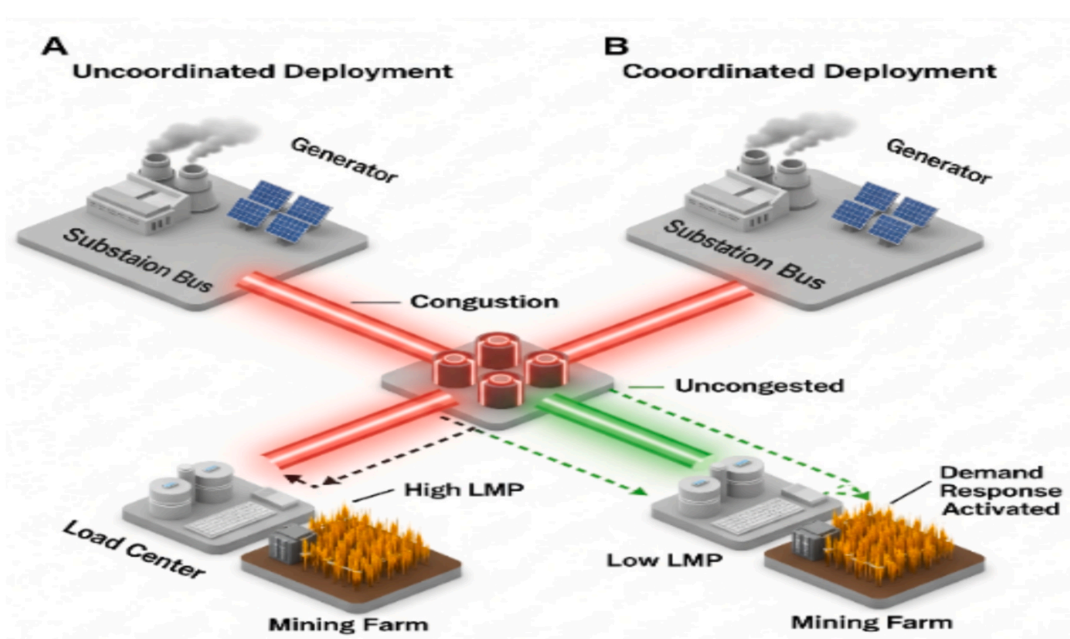

Fig. 11. A comparison of uncoordinated vs. coordinated deployment of mining farms at the transmission level. The uncoordinated scenario (A) leads to line congestion and high LMP, while the coordinated scenario (B) utilizes demand response to alleviate congestion.

Table 11
A critical review of research on the power quality and infrastructure impacts of cryptocurrency mining on distribution networks.

Ref.	Main research objective	Type of ancillary service	Methodology/ model	Tools and data	Key findings	Main contribution	Limitations and technical remarks
[112]	Comprehensive assessment of the impact of high penetration of home miners on distribution network planning and operation	Simultaneous demand (ADMD), transformer aging, and power quality	Technology Acceptance Model (TAM), ADMD modeling, IEEE-based thermal aging analysis, power quality measurement	Case study of Tehran distribution network; load data from 40 substations; measurements from Antminer **S9j**	**5 %** penetration increased peak demand by **26.81%** and transformer aging rate by **21** times; current THD and third harmonic exceeded standards	A direct model linking miner penetration percentage to transformer aging rate; a practical risk assessment tool for distribution operators	Focused on older miner models; assumes constant mining load; results depend on Iran's specific grid characteristics
[113]	Analysis of load and power quality in a mining center connected to a distribution system	Harmonics, power factor, and load characteristics	Real-world **3**-day measurement (**15**-min intervals) using a power quality analyzer; compared to IEEE Std. **519**	**10** Antminer S9 units; ION **7650** analyzer; three-feeder distribution network model	Miners showed leading power factor near unity; current THD of **9.04 %** exceeded IEEE standard limit (**5 %);** transformer connection filtered triplen harmonics	One of the first empirical analyses of a miner set showing leading PF and the harmonic mitigation role of transformer configuration	Limited to **10** devices; extrapolated impacts on larger networks are inferred, not directly measured
[114]	Evaluation of a single mining device's impact on low-voltage distribution network power quality, compared to a computer cluster	Harmonics, power factor, and load profile	Continuous week-long power quality monitoring; comparative analysis	One 1000 W mining rig (6 GPUs); one 42-PC cluster; measurements at University of Novi Sad's LV network (Serbia)	Mining rig showed better power factor (0.988) and lower THDI (**9.04 %)** than the cluster (**31.7 %)** but higher and more stable load	First experimental comparison between a custom mining rig and a compute cluster; demonstrated the effect of PSU quality on power quality per device	Only one custom rig studied; results may differ for ASIC miners; conclusions on large-scale penetration are inferred
[115]	Development of an optimization framework to improve demand response (DR) efficiency in microgrids and prevent rebound peaks	Coordinated DR, peak shaving, demand-side flexibility	Two-stage multi-objective optimization with lexicographic approach: Stage 1 minimizes operational cost; Stage 2 minimizes peak load	Standard test system with three interconnected microgrids including DERs, batteries, and flexible loads	Coordinated model reduced peak by 12.96 % and improved load factor by 15.17 %; uncoordinated model increased peak by **5.49 %;** operational cost remained unchanged	A two-stage framework separating economic optimization from flexibility enhancement; introduced a new flexibility index (APFDPPI)	Purely simulation-based study; not directly applied to mining loads; practical issues such as communication and control not addressed

loads.

Real-world implementation barriers: latency, standardization, and stability

"While the technical potential of miners as ancillary service providers is clear (as discussed in Sections 4.2.1–4.2.3), their widespread integration is currently hindered by significant technical and regulatory constraints.

a) Latency and Communication: A primary barrier is latency. While the device latency of an ASIC is near-instantaneous (milliseconds), the communication latency is the bottleneck. A signal from a TSO must travel to an aggregator, be processed, and then be dispatched to thousands of individual miners. For high-value services like Fast Frequency Response (FFR), which require a response in under two seconds (or even sub-second), this entire communication chain must be exceptionally robust, secure, and fast [115]. This requires a high-availability, two-way communication infrastructure (e.g., Fiber optics) that is often not present at remote mining locations.

b) Standardization and Interoperability: Progress on standardized protocols has been slow. The mining ecosystem is highly fragmented with proprietary APIs for hardware, firmware (e.g., Braiins OS), and mining pool management. This lack of interoperability makes it difficult for aggregators and TSOs to adopt a "plug-and-play" solution [117]. While industry standards for demand response (like OpenADR, IEEE 2030.5, or IEC 61850) exist, they are not natively supported by mining equipment, requiring costly custom-built "gateway" solutions by aggregators [115].

c) Cybersecurity Risks: The aggregation of flexible loads into a Virtual Power Plant (VPP) creates a significant systemic risk. A malicious actor gaining control of an aggregator's platform could trigger a "coordinated load-dropping attack," simultaneously switching off gigawatts of load. Such an event would create a severe frequency disturbance, potentially leading to cascading failures [115]. This risk makes TSOs and regulators highly cautious, demanding stringent cybersecurity protocols (e.g., NERC CIP standards) that the mining industry is only beginning to address [115].

d) Stability in Weak Grids (Low-Inertia): The interaction with IBR-dominated weak grids is a dual-edged sword. Uncoordinated, volatile mining loads can exacerbate instability in these low-inertia systems, which already suffer from high Rate of Change of Frequency (RoCoF) [115]. However, if properly controlled, these same miners are an ideal resource to solve this problem. Their ultra-fast curtailment capability can provide "Synthetic Inertia" or FFR, acting as a "virtual brake" to arrest frequency deviations far faster than conventional generators, thus enhancing stability [115]."

4.3. Environmental footprint of cryptocurrency mining: quantification, mitigation strategies, and sustainability pathways.

As discussed in the previous section, the exceptionally high energy consumption required by the PoW algorithm, which underpins prominent cryptocurrencies such as Bitcoin, has placed the industry at the center of significant debates concerning environmental sustainability. These environmental concerns are not limited to electricity use but also encompass broader dimensions, including carbon emissions, water consumption, and the generation of electronic waste (e waste). Such impacts are considered to be in direct opposition to global objectives for greenhouse gas reduction, as outlined in international agreements such as the Kyoto Protocol and the Paris Agreement, which aim to transition nations toward a low carbon economy [118–120].

Nonetheless, the unique characteristics of the industry, particularly the operational flexibility of mining loads, provide potential avenues for innovative solutions along the path toward clean energy. Accordingly, this section presents an analytical review of the literature, first quantifying the multiple dimensions of these environmental challenges and subsequently examining proposed mitigation strategies aimed at minimizing their impact.

Holistic quantification of energy, carbon, and water footprints in cryptocurrency mining ecosystems

To fully understand the environmental impacts of cryptocurrency mining, it is essential to quantitatively assess its multiple dimensions. Research in this field has focused on two primary aspects:

A) Energy Consumption, Carbon, and Water Footprint

Numerous studies have analyzed the electricity usage and associated greenhouse gas emissions of mining operations. Stoll et al. [121], using data from hardware manufacturers and IP address geolocation, estimated that by November 2018, the Bitcoin network consumed approximately 45.8 TWh of electricity annually, generating between 22 and 22.9 million tons of CO_2 emissions. This footprint is comparable to the total carbon emissions of countries such as Jordan or Sri Lanka. A more recent study by Laimon et al. [122], employing a comprehensive system dynamics approach, projected substantially higher values for 2023, with electricity consumption reaching 119.7 TWh and a water footprint of 1,859 million m^3, revealing a hidden environmental crisis. These analyses extend beyond Bitcoin; Li et al. [123] demonstrated through practical experiments that other proof of work cryptocurrencies, including Monero, also exhibit significant energy consumption and carbon footprints.

B) Electronic Waste (E waste)

Another significant but often overlooked environmental concern is the large volume of e waste generated by mining hardware. De Vries et al. [81] developed a methodology to estimate the "economic lifespan" of ASIC devices, finding that the average operational life of a Bitcoin mining unit is only 1.29 years. As a result, the Bitcoin network produces roughly 30.7 kilotons of e-waste annually, comparable to the IT equipment waste of an entire country, such as the Netherlands. This finding underscores that the environmental impact of cryptocurrency mining extends beyond carbon emissions to include the full lifecycle of its hardware.

While unmanaged mining has been criticized for its carbon and electronic waste footprint, the environmental impact of managed, grid-interactive mining depends largely on the regional electricity mix and hardware management practices. When miners operate as controllable loads that absorb surplus renewable generation, their marginal CO_2 emissions can approach zero, effectively reducing system-level curtailment and displacing fossil-based peaking generation. In contrast, continuous operation in carbon-intensive grids amplifies emissions in proportion to the local grid's carbon intensity. Compared with other flexible resources, such as battery storage, mining devices exhibit lower embodied emissions but generate a distinct e-waste stream due to short ASIC lifetimes. Addressing this requires adoption of circular-economy practices, including refurbishment, component reuse, and regulated recycling, similar to the end-of-life pathways being developed for lithium-ion batteries. Therefore, from a lifecycle perspective, the sustainability of mining as a flexibility resource is highly context-dependent, improving markedly in low-carbon or renewable-rich regions and under strict hardware-recycling and operational-dispatch controls.

Innovative pathways for sustainable cryptocurrency mining: advanced mitigation strategies and consensus mechanism alternatives

A review of recent literature identifies three main pathways that researchers and industry professionals widely explore to reduce the environmental impact of cryptocurrency mining.

A) Integration of renewable energy sources into cryptocurrency mining Operations

The most straightforward and immediate solution is to power mining facilities with clean energy. Several studies confirm that this approach is both environmentally and economically viable. For example, Hakimi et al. [124]conducted a case study in the UAE and found that building a solar power plant for a mining center avoided about 50,000 tons of CO_2 emissions each year. In addition, it reduced the investment payback period from 8.1 years to 3.5 years compared to selling the electricity to the grid. Similarly, Siddique et al. [125] compared various renewable energy sources and showed that wind and hydropower not only provide low-carbon alternatives but also offer highly profitable solutions for cryptocurrency mining operations.

B) Leveraging cryptocurrency mining as a decarbonization and grid flexibility tool

Beyond merely sourcing clean energy, cryptocurrency mining can actively support the decarbonization process and enhance grid flexibility. Goodkind et al. [110] identified three critical roles for mining in the energy transition: (1) providing demand response services, (2) consuming surplus renewable energy that would otherwise be curtailed, and (3) reducing methane emissions through the utilization of flare gas. Expanding on this concept, another study [126] proposed a synergistic model in which a mining facility is co-located with a solar power plant and a green hydrogen production unit. By consuming excess solar energy, the mining facility prevents energy waste and generates an additional revenue stream, significantly reducing the cost of green hydrogen production. These approaches demonstrate that mining operations can be transformed from environmental liabilities into integral components of a clean energy solution.

C) Algorithmic and technological innovations for sustainable consensus mechanisms

From a long term sustainability perspective, many researchers argue that the ultimate solution lies in replacing, rather than merely optimizing, the Proof of Work consensus mechanism. De Vries et al. [119] and Laimon et al. [122] emphasize that as long as PoW remains dominant, the issues of high energy consumption and electronic waste cannot be fully addressed. Promising alternatives include PoS, adopted by networks such as Ethereum, which reduces energy consumption by over 99 %, and other consensus protocols, such as the Stellar Consensus Protocol. Although transitioning to these mechanisms presents technical and organizational challenges, it represents the most robust pathway to achieving a truly sustainable cryptocurrency ecosystem.

Collectively, these three pathways form a comprehensive, multi-layered framework for mitigating the environmental footprint of cryptocurrency mining. Integration with renewable energy sources enables immediate reductions in emissions, while decarbonization oriented models leverage the operational flexibility of mining to provide broader benefits to the energy system. Simultaneously, alternative consensus mechanisms address the fundamental drivers of excessive energy consumption and electronic waste. When implemented in a coordinated and strategic manner, these approaches offer a robust and complementary strategy that aligns cryptocurrency mining with global sustainability targets and energy transition objectives.

Real world implementations of sustainable cryptocurrency mining: industrial-scale projects and cross-sector synergies

The strategies and solutions outlined in the academic literature have increasingly transitioned from theory to practice in recent years, materializing as innovative, industrial scale projects worldwide. These initiatives illustrate how thoughtful design and creative implementation can transform the environmental challenges of cryptocurrency mining into economic and technical opportunities. Table 12 summarizes some of the most prominent projects in this domain.

A combined review of scholarly studies and practical implementations highlights the evolution of the mining industry from a standalone, high consumption activity into an integrated, synergistic component of the broader energy ecosystem. Projects such as MARA and Soluna in Texas exemplify the practical realization of concepts proposed in studies like [110]. Similarly, European projects, including those undertaken by Marathon and Tera-hash in Finland, have successfully operationalized cross sector synergies, such as district heating systems, which were previously explored theoretically in studies like [126]. The analysis of these projects indicates that, through intelligent system design, cryptocurrency mining can not only mitigate its environmental footprint but also serve as a strategic tool to enhance overall energy efficiency and foster sustainability across other industrial sectors.

4.4. Policy and regulatory frameworks influencing the sustainable development of cryptocurrency mining

The sustainability and profitability of the cryptocurrency mining industry are more closely linked to energy policies and electricity market structures than to the short-term price volatility of digital assets. For instance, Majumder et al. [103] demonstrated that miners adjust their behaviour based on electricity prices and grid tariffs, strategically reducing their consumption to avoid the high costs imposed by distribution and transmission policies.

In this context, de Vries et al. [132] highlight the attractiveness of countries such as Iran, Russia, and China for mining due to government subsidies and low electricity pricing. While these policies create a significant competitive advantage, they also offer the potential to use mining to enhance the profitability of renewable energy projects [133]. This synergy presents a strategic opportunity for these nations to simultaneously advance environmental objectives and support an emerging economic industry. Therefore, the future of cryptocurrency mining depends on the informed decisions of policymakers.

Countries that establish transparent regulations to leverage the flexibility of miners for grid stability, while reinforcing the synergy between mining and clean energy, can transform this industry into a driver of sustainable, eco friendly economic growth. Ultimately, policy and regulatory frameworks can function as both catalysts and barriers to sustainable cryptocurrency mining. Coordinated policies that integrate energy, environmental, and digital asset regulations will be essential to harness the potential of mining while mitigating its adverse impacts.

The challenge of policy alignment is best illustrated by a global geographic analysis, as presented in Fig. 12. This figure provides a global match analysis, comparing cryptocurrency mining hotspots (based on 2024–2025 CCAF data [134], Fig. 12-A) against high-potential renewable energy zones (exemplified by solar irradiance GHI, Fig. 12-B).

This comparison reveals three distinct national archetypes: (1) “High-Match” zones (e.g., the USA), where supportive policy aligns high hash rate with abundant renewables; (2) “Infrastructure Mismatch” zones (e.g., North Africa), where massive renewable potential is untapped due to grid and infrastructure deficits; and (3) “Policy Mismatch” zones.

This third archetype, most clearly represented by China, illustrates the core “policy paradox.” Although the country possesses some of the world’s most abundant renewable resources particularly hydropower in Sichuan and Yunnan the implementation of a stringent mining ban in 2021 aimed at curbing energy consumption and financial speculation produced an unintended effect. Reportedly, renewable energy curtailment increased because miners were no longer available to act as a “buyer of last resort” for surplus hydropower [136].

This paradox highlights a critical system optimization failure and raises the question: how can the VESS concept be promoted in regions with high renewable potential but restrictive policies? The solution lies in policy innovation to reconcile these goals. We propose three complementary pathways:

1. Re-framing as Technical Grid Assets: A key solution is to decouple the technical function from the economic output. This is not merely semantic but a technical reclassification. Policymakers could re-label these facilities not as “cryptocurrency mining” (which carries political stigma) but as “Interruptible Data Centers” or “Flexible Computational Loads”. This allows regulators to integrate them into established grid service mechanisms, such as Demand Response (DR) markets, providing non-spinning reserves, or ancillary service procurement. The focus shifts from the politically sensitive economic output to the verifiably valuable grid-balancing function [137]
2. Structuring ’Behind-the-Meter’ Regulatory Sandboxes: Governments could create “regulatory sandboxes” or special economic zones. In these zones, mining would be permitted only under strict conditions: (1) proof of 100 % renewable energy use (often “behind-the-meter”), (2) mandatory participation in grid ancillary services upon request, and (3) adherence to e-waste regulations [137]. To incentivize participation, these zones could offer preferential electricity tariffs (e.g., zero-cost for verifiably curtailed energy) or fast-tracked permitting for co-located renewable projects. Success would require smart metering and third-party auditing for continuous compliance verification

Table 12
A summary of prominent real-world projects implementing sustainable cryptocurrency mining solutions.

Project/company	Location	Capacity (MW)	Energy source/carbon reduction method	CO_2 reduction/system benefits	Power/grid services	Project year	Ref
MARA – Texas wind farm	Hansford County, Texas	**114** MW wind power	Mining **~30 %** of time using wind power (curtailment)	Near zero carbon during operation; using surplus grid power	Load balancing during high wind periods	**2025**	[127]
MARA + NGON – associated gas	US Oil Fields	**25 MW**	Using flare gas for power generation	Reduced **29,300** tons CO_2 in **5** months; equivalent to removing **6,800** gasoline cars	Off-grid; grid-independent, reducing peak load pressure	**2024**	[128]
Soluna – Project Dorothy 2	Silverton, Texas	**48 MW**	PPA with EDF for wind power	Prevents wind curtailment; **2.2** EH/s hash rate	Flexible mining, load-following during low-price hours	**2024–2025**	[129]
Terahash – Genesis/ Aurora (Finland)	Finland	**1 MW**	**100 %** renewable power (wind & solar) + heat recovery	**90 %** heat energy recovery; reuse in district heating	Connected to district heating, fast demand response	**2024**	[130]
Marathon – District Heating (Finland)	Satakunta, Finland	**2 MW**	District heating from mining heat	Heating for over **11,000** people; replaces natural gas	Connected to district heating grid; municipal collaboration	**2024**	[84]
Braiins – Hash-heating Europe	Europe	Modular (up to 1 MW)	Mining heat transferred to building and industrial heating	Saves up to **1,500** tons CO_2 per MW per year	Direct connection to district/home heating systems	**2025**	[131]

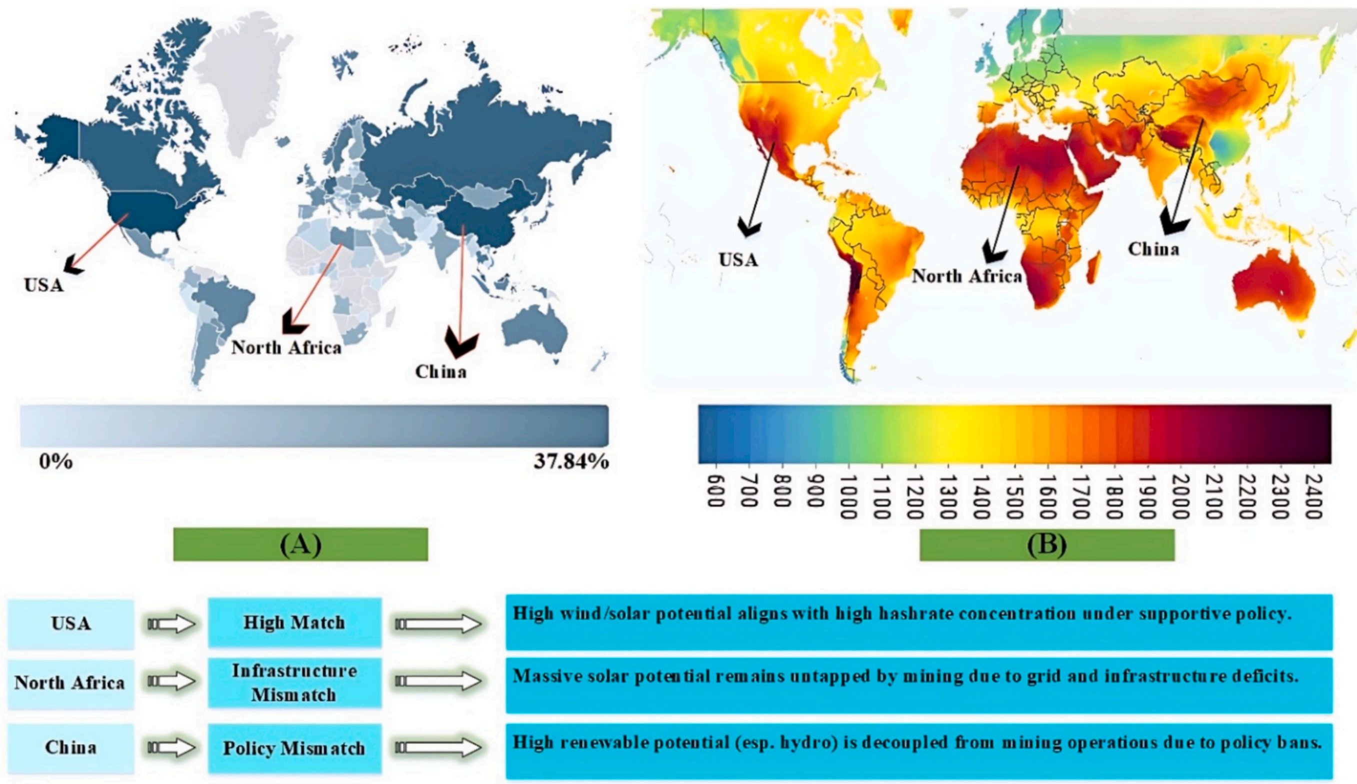

Fig. 12. Global Match Analysis of Cryptocurrency Mining Hotspots and Renewable Energy Potential. (A) Global hashrate distribution by country (based on 2024–2025 CCAF data [134]). (B) Global Solar Potential map (GHI in kWh/m^2). The analysis below the maps illustrates the three key alignment scenarios discussed in Section 4.4. (Data Sources: CCAF [135], Global Solar Atlas).

3. Creating Market-Based Instruments for Curtailment: A more advanced solution is to decouple these operations from general power markets and create a secondary market. We propose the creation of a "Green Curtailment Certificate" (GCC). Miners would earn these tradable certificates only when verifiably consuming energy that would have otherwise been curtailed by the grid. This creates a precise, market-driven incentive to utilize waste energy, bypassing the political complexities of supporting "mining" itself

These policy strategies, by focusing on renewable optimization and grid stability rather than promoting mining per se, offer a viable path to reconciling the seemingly contradictory goals of energy, economics, and environmental policy.

Future perspectives and research gaps

The integration of cryptocurrency mining into power systems is recognized as a promising yet nascent research domain. The analysis presented in this review highlights that, while substantial opportunities exist, a coordinated, multidisciplinary research effort is required for these benefits to be realized. This section systematically outlines these gaps, first by building on the limitations of existing studies and then by proposing novel interdisciplinary research directions.

Cited research gaps: building on existing work

Grid Planning and Operations: A crucial short-term analysis of transmission congestion was provided by Menati et al. [111]; however, significant gaps remain in long term strategic planning. Future research should address questions such as: How can optimal siting and sizing models for cryptocurrency mining facilities be integrated into Transmission Expansion Planning (TEP) so that future grid bottlenecks are minimized? Likewise, while profit optimization models for mining operations were pioneered by Menati et al. [107], idealized conditions were assumed in their analyses. The next step is to ensure that dynamic control models are developed that account for real world factors, including communication latency, strategic market bidding, and cybersecurity threats, so that the reliability, resilience, and operational impact of these resources can be accurately evaluated.

Distribution Network Management: Foundational work on the accelerated aging of transformers caused by uncoordinated mining activity was conducted by Hajiaghapour Moghimi et al. [112]. While valuable diagnostics were provided by these studies, a clear need for proactive solutions is revealed. Future efforts should focus on ensuring that active control frameworks for Distribution System Operators (DSOs) are designed and validated so that residential mining fleets can be managed, equipment strain is mitigated, collective flexibility is harnessed, and data privacy and cybersecurity compliance are maintained.

Synthesized research frontiers: proposing novel directions

Market Design and Financial Hedging: A central challenge is posed by the inherent tension between the need for reliable grid services and the volatility of cryptocurrency markets. This raises a key research question: What innovative market products or financial instruments such as long-term, fixed price contracts for ancillary services, potentially hedged against cryptocurrency futures can be developed to decouple physical grid services from asset volatility, thereby ensuring that reliability for system operators is maintained and predictable revenue streams for miners are secured? Addressing this question will require that close collaboration among academic researchers, industry stakeholders, and regulatory authorities be established so that practical market mechanisms can be designed and implemented.

AI-Driven Decentralized Control: The inherently decentralized nature of cryptocurrency mining highlights the need for advanced, scalable control paradigms. Future research should ensure that AI driven, decentralized platforms potentially leveraging reinforcement learning and federated learning techniques are developed. The fundamental question is: How can thousands of independent miners be orchestrated into a cohesive Virtual Power Plant (VPP) that delivers reliable grid services without overloading local distribution networks, while simultaneously ensuring that privacy and cybersecurity protections are maintained?.

Circular Economy for Mining Hardware: The e waste problem, as emphasized by de Vries et al. [81], is recognized as a major sustainability challenge for the industry. This challenge opens a research frontier centered on circular economy principles. A critical question is: What is the techno economic feasibility of establishing a “second life” market in which aging or less efficient ASIC devices are repurposed for lower value grid services such as interruptible loads in regions with frequent renewable energy surplus so that their operational lifespan is extended and e waste is reduced?.

To complement theoretical advances, large scale field trials and pilot projects are considered essential for validating models under real world operational conditions. Collaboration among academic institutions, cryptocurrency mining operators, grid operators, and policymakers is required to ensure that proposed solutions are made practical, scalable, and aligned with regulatory frameworks. Through such collaborative efforts, the integration of mining operations into ancillary service provision can be accelerated, while system reliability and efficiency are maintained.

Socio Economic and Environmental Impacts: Future research is expected to be approached from a holistic perspective, with socio economic and environmental considerations incorporated at both local and regional scales. Potential benefits, such as job creation and economic growth, should be evaluated alongside challenges like increased energy demand, water consumption, and community acceptance. A structured framework is provided in Table 13, where key research questions, suggested methodologies, and anticipated contributions to both energy system resilience and sustainable development goals are detailed.

Conclusion

In this comprehensive review, the integration of cryptocurrency mining operations within modern power systems has been systematically examined, with technical, economic, environmental, and policy dimensions being addressed. Drawing on over 120 peer reviewed publications from 2019 to 2025, the first integrated framework for assessing cryptocurrency mining as both a grid challenge and a potential asset has been presented.

From a technical perspective, distinctive characteristics of mining operations have been identified, setting them apart from conventional industrial loads. The ability to disconnect instantaneously, operate independently of geographical constraints, and respond rapidly to grid signals has been shown to enable effective participation in ancillary service provision, including frequency regulation, voltage support, and operational reserves. When coordinated with renewable generation, mining has been demonstrated to function as a form of “economic storage,” through which surplus energy is absorbed during peak production periods and renewable curtailment is mitigated without reliance on physical storage systems.

It has been emphasized throughout the review that cryptocurrency mining represents a unique load category. Sub-second disconnection capabilities, high geographical flexibility, and controllability factors exceeding 95 % have been reported as key enabling features. While these capabilities offer potential system support, it has been cautioned that uncoordinated deployment introduces substantial risks. Distribution transformer aging can be accelerated by up to 21 times, and approximately 30.7 kilotons of electronic waste are generated annually, given an average hardware lifespan of just 1.29 years.

Conversely, measurable benefits have been observed when coordinated integration is implemented. Within grid connected microgrids, operational cost reductions of up to 46.5 % have been demonstrated under the Cryptocurrency Energy Storage System (CESS) framework, while annualized cost reductions of 68.1 % for distributed solar generation have been achieved. Mining profitability has been shown to be increased by over 20 % through participation in ancillary service markets, and solar project payback periods have been reduced from 8.1 to 3.5 years when mining is coupled with renewable resources. In addition, annual CO_2 emission reductions on the order of tens of thousands of tons have been facilitated through this coordinated approach.

Collectively, these findings are indicative of the dual role of cryptocurrency mining: it is regarded both as a potential system burden and as a flexible, high value resource. To ensure that its benefits are realized, coordinated planning, the development of active control frameworks, and the strategic alignment of mining operations with grid objectives and sustainability goals are required.

From an economic standpoint, cryptocurrency mining is estimated to constitute a 15–20 billion annual electricity market. However, profitability is recognized as remaining highly sensitive to cryptocurrency price volatility, with fluctuations by factors of three to five being observed across market cycles. From an environmental perspective, annual electricity consumption of 100–130 TWh has been reported, comparable to that of medium sized economies, while CO_2 emissions in the range of 48.6–64.4 million tons per year are generated.

The integration of mining operations into power systems is recognized as necessitating fundamental advancements in grid management capabilities, as enhanced control architectures are expected to be

Table13
A systematic framework for future research directions.

Research theme	Key research question	Proposed methodological approach	Expected impact & contribution
Grid planning & operations	How can optimal siting of mining facilities be integrated into long-term Transmission Expansion Planning (TEP)?	Advanced optimization (e.g., Multi-Stage Stochastic Optimization) integrated with TEP models.	A framework for co-planning grid and mining infrastructure to prevent future congestion.
Market design & economics	What financial instruments can hedge against cryptocurrency volatility to ensure reliable grid service provision?	Financial Engineering; Techno-Economic Modeling; Contract Theory.	Decouples physical reliability from market volatility, enabling stable, long-term contracts.
AI, control & automation	How can AI create a decentralized Virtual Power Plant (VPP) from thousands of independent miners?	Reinforcement Learning; Federated Learning; Agent-Based Modeling.	Enables mass participation of small-scale, distributed assets in grid services while ensuring cybersecurity and privacy.
Sustainability & circular economy	What is the techno-economic case for “second-life” ASICs to mitigate e-waste?	Life-Cycle Assessment (LCA); Cost-Benefit Analysis; Hardware testing.	A circular economy model to reduce the lifecycle environmental footprint of mining hardware.
Field trials & collaboration	How can multi-stakeholder pilot projects validate and scale proposed solutions in real-world conditions?	Pilot Projects; Collaborative Frameworks; Regulatory Analysis.	Practical validation of theoretical models; ensures regulatory and market alignment.
Socio-economic & environmental	What are the broader socio-economic and environmental impacts of integrating mining into power systems?	Impact Assessment; Social Surveys; Environmental Modeling.	Comprehensive understanding of community, economic, and environmental outcomes.

adopted by Distribution System Operators (DSOs) to effectively manage these highly flexible loads. Among the key technical requirements, the development of grid aware ASICs and AI driven control platforms is particularly essential, since they enable real time optimization that supports grid stability, economic efficiency, and environmental performance.

Policy frameworks must evolve to recognize mining operations in their dual role as both energy consumers and potential grid assets, with dynamic tariff structures supporting flexible loads. Moreover, innovative market mechanisms should enable these facilities to participate in local grid services, while comprehensive regulatory approaches are essential to balance economic opportunities with environmental and stability concerns, ensuring sustainable integration.

Despite these advancements, several critical research gaps remain to be addressed. These include developing optimal siting methodologies, which account for grid congestion, renewable resource availability, and environmental constraints (1), as well as designing integrated control systems that enable coordinated operation of mining facilities, renewable generation, and energy storage (2). In addition, comprehensive lifecycle assessment frameworks are needed to consider both direct mining impacts and indirect grid benefits (3), while financial instruments should be created to manage cryptocurrency market volatility alongside grid related uncertainties (4). Finally, transition pathways toward low energy consensus mechanisms, including Proof of Stake and emerging Proof of Useful Work (PoUW) concepts, must be explored (5).

The inherent limitations of this review are also acknowledged, particularly given the rapidly evolving nature of both cryptocurrency technologies and power system infrastructures. Moreover, the limited availability of real world operational data from mining facilities, alongside regional variations in regulatory and market structures, further constrains the generalizability of the findings. While the analysis has primarily focused on Proof of Work systems, coverage of alternative consensus mechanisms remains limited, highlighting areas for future investigation.

The integration of cryptocurrency mining into contemporary power systems embodies both pressing challenges and transformative opportunities. Its future trajectory will hinge on coordinated engagement among engineers, policymakers, system operators, and industry stakeholders to exploit operational flexibility while constraining infrastructural vulnerabilities and environmental risks. Realizing this vision requires a paradigm shift: mining must be reconceptualized from an energy intensive liability into a controllable and responsive grid resource that facilitates renewable integration and enhances overall system resilience. Ultimately, the contribution of cryptocurrency mining to power system decarbonization and modernization will depend on the establishment of technically robust, economically viable, and environmentally sustainable integration pathways. This review seeks to advance such efforts by providing a multidisciplinary foundation for informed decision-making at the nexus of digital innovation and electric power systems.

CRediT authorship contribution statement

Alireza Jafari: Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Taher Niknam:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Conceptualization. **Ali Taghavi:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Investigation, Conceptualization. **Sadreddin Saleh:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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